

Decentralized Online Learning with Hard Constraints: A Doubly-Bounded Queue Approach

Yituo Liu, *Student Member, IEEE*, Weiyi Qin, *Student Member, IEEE*, Wei Bao, *Member, IEEE*,
Juncheng Wang, *Member, IEEE*, Jianxiong Guo, *Member, IEEE*, and Min Zhou

Abstract—We consider decentralized online convex optimization with time-varying constraints and conduct performance analysis using two stringent metrics: network dynamic regret with respect to the online global solution benchmark, and hard constraint violation that does not allow any compensated violation over time and learners. We propose an efficient algorithm called Constrained Online Learning with Doubly-bounded Queue (COLDQ), which introduces a novel virtual queue that is both lower and upper bounded, allowing tight control of the constraint violation without needing the Slater’s condition. We prove via a new Lyapunov drift analysis that COLDQ provides $\mathcal{O}(T^{\frac{1+V_x}{2}})$ network dynamic regret and $\mathcal{O}(\min\{T^{\frac{3+V_x}{4}}, T^{V_g}\})$ hard constraint violation, where V_x and V_g capture the dynamics of the loss and constraint functions. For the first time, the two bounds smoothly approach to the best-known $\mathcal{O}(T^{\frac{1}{2}})$ regret and $\mathcal{O}(1)$ violation, as the dynamics of the losses and constraints diminish, under both centralized and decentralized settings. For strongly convex loss functions, COLDQ provides $\mathcal{O}(\log T)$ static regret and $\mathcal{O}(\min\{T^{\frac{1}{2}}(\log T)^{\frac{1}{2}}, T^{V_g}\})$ hard constraint violation. Simulation results based on synthetic and canonical datasets demonstrate that COLDQ substantially outperforms the state-of-the-art approaches for both convex and non-convex loss functions.

Index Terms—Decentralized online learning, constrained optimization, virtual queue, dynamic regret, constraint violation.

I. INTRODUCTION

In many online learning applications, optimization losses and constraints are dynamic over time. Online Convex Optimization (OCO) [2], [3], as the intersection of learning, optimization, and game, is a vital framework for solving online learning problems under uncertainty. It has broad applications in networking, such as advertisement placement [4], load balancing [5], network virtualization [6], and task scheduling [7].

In the standard OCO setting [8]–[17], a learner selects online decisions from a known convex set to minimize a sequence of time-varying convex loss functions. The information of

each loss function, however, is only revealed to the learner after the decision has been made. Given this lack of current information, the objective of the learner becomes to minimize the *regret*, which is the accumulated difference between the losses incurred by their online decisions and those of some benchmark solutions. The seminal OCO work [8] considered both the *static* regret to an offline benchmark and the *dynamic* regret to an online benchmark. The proposed online projected gradient descent algorithm provided a dynamic regret bound that *smoothly* approaches to $\mathcal{O}(T^{\frac{1}{2}})$ static regret, as the accumulated variation of the loss functions reduces, *i.e.*, the OCO algorithm keeps pace with the dynamics of the losses.

The projection operation to strictly satisfy the constraints at each time can incur heavy computation. Furthermore, in many applications, *e.g.*, demand response [18], smart grid [19], and energy harvesting [20], the online decisions are subject to constraints that are allowed to be violated at certain time slots [21]–[54]. OCO with *soft* constraint violation that measures the amount of compensated violations over time was first considered in [21]. In contrast, with a goal to limit the instantaneous violation, a stronger notion of *hard* constraint violation that does not allow any compensated violation over time was introduced in [25].

For OCO with *fixed* constraints [21]–[27], the best-known soft and hard constraint violation bounds are both $\mathcal{O}(1)$ [23], [27]. For OCO with *time-varying* constraints [28]–[36], most existing works focused on the static regret [28]–[32]. Dynamic regret for time-varying constrained OCO was more recently studied in [33]–[37]. As the accumulated variation of the constraint functions reduces, the best-known soft and hard constraint violation bounds for time-varying constrained OCO approach to $\mathcal{O}(T^{\frac{1}{2}})$ [36] and $\mathcal{O}(T^{\frac{1}{2}} \log T)$ [28], respectively. However, none of the constraint violation bound recovers the best-known $\mathcal{O}(1)$ violation for fixed constraints, *i.e.*, the constrained OCO algorithms do *not* keep pace with the dynamics of the constraints.

While all the aforementioned works focus on *centralized* OCO, modern large-scale machine learning applications often involve heavy computation that can easily overwhelm centralized systems. *Decentralized* OCO [38]–[54] alleviates the burden on a single learner by distributing the computational tasks among multiple learners. In decentralized OCO, each learner only has access to its own local loss function as one part of the global loss function. All the learners collaboratively optimize their local decisions to minimize the *network* regret on the global loss, through peer-to-peer communication over decentralized networks. This can further remove the communication bottleneck incurred by the central server in wireless

Y. Liu, W. Qin, and J. Wang are with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China (e-mail: csytliu@comp.hkbu.edu.hk; cswyqin@comp.hkbu.edu.hk; jcwang@hkbu.edu.hk).

W. Bao is with the School of Computer Science, The University of Sydney, Sydney, Australia (e-mail: wei.bao@sydney.edu.au).

J. Guo is with the Advanced Institute of Natural Sciences, Beijing Normal University, Zhuhai, China, and also with the Guangdong Key Lab of AI and Multi-Modal Data Processing, Beijing Normal-Hong Kong Baptist University, Zhuhai, China (e-mail: jianxiongguo@bnu.edu.cn).

M. Zhou is with Huawei Technologies Co., Ltd., Shenzhen, China (e-mail: zhoumin27@huawei.com).

Y. Liu and W. Qin contributed equally to this work.

A preliminary version of this work was presented in the AAAI Conference on Artificial Intelligence (AAAI), 2025 [1].

online federated learning over both digital [55] and analog communication networks [56].

Most existing works on decentralized constrained OCO focus on the *soft* constraint violation [45]-[51]. Among them, static regret for decentralized OCO with *fixed* constraints was studied in [45]-[47]. Decentralized OCO with *time-varying* constraints was studied in [48]-[51]. The current best soft constraint violation bounds for both fixed and time-varying constraints are $\mathcal{O}(T^{\frac{3}{4}})$ [47], [48]. Only a few works studied decentralized constrained OCO with *hard* constraint violation [52]-[54]. The current best hard constraint violation bounds for both fixed and time-varying constraints are also $\mathcal{O}(T^{\frac{3}{4}})$ [53], [54]. Similar to centralized constrained OCO, none of the constraint violation bound recovers the best-known $\mathcal{O}(1)$ violation for fixed constraints, *i.e.*, existing constrained OCO algorithms do *not* keep pace with the dynamics of the constraints under *both* centralized and decentralized settings.

The above discrepancies motivate us to pose the following key question: *Can a decentralized constrained OCO algorithm provide a network dynamic regret bound and a hard constraint violation bound that smoothly approach to the best-known $\mathcal{O}(T^{\frac{1}{2}})$ regret and $\mathcal{O}(1)$ violation, respectively, as the dynamics of the losses and constraints diminish?* Our answer is yes.

Contributions: We summarize our main contributions below.

- We propose an effective algorithm named Constrained Online Learning with Doubly-bounded Queue (COLDQ) for tackling decentralized OCO problems with time-varying constraints. Existing virtual-queue-based approaches rely on either a lower or an upper bound of the virtual queue to control the constraint violation. In contrast, we introduce a novel virtual queue that enforces both a lower and an upper bound to strictly control the hard constraint violation, even without the commonly assumed Slater's condition.
- We analyze the performance of COLDQ via a new penalized Lyapunov drift design that leverages both the lower and upper bounds of the virtual queue. We show that COLDQ provides $\mathcal{O}(T^{\frac{1+V_x}{2}})$ network dynamic regret and $\mathcal{O}(\min\{T^{\frac{3+V_x}{4}}, T^{V_g}\})$ hard constraint violation, where V_x and V_g capture the dynamics of the losses and constraints (see their definitions in (17) and (18)). For the first time, the two bounds smoothly approach to the best-known $\mathcal{O}(T^{\frac{1}{2}})$ regret and $\mathcal{O}(1)$ violation as $V_x \rightarrow 0$ and $V_g \rightarrow 0$, under both centralized and decentralized settings.
- When the loss functions are strongly convex, we show that COLDQ provides $\mathcal{O}(\log T)$ static regret and $\mathcal{O}(\min\{T^{\frac{1}{2}}(\log T)^{\frac{1}{2}}, T^{V_g}\})$ hard constraint violation. Still, for the first time, the constraint violation bound smoothly approach to $\mathcal{O}(1)$ as $V_g \rightarrow 0$, under both centralized and decentralized settings.
- We conduct experiments to evaluate the performance of COLDQ on both time-varying and fixed constraints using synthetic datasets. Furthermore, we apply COLDQ to practical communication-efficient decentralized learning with both convex and non-convex losses using canonical

datasets. Numerical results confirm the effectiveness of COLDQ over the state-of-the-art approaches.

Organizations: The rest of this paper is organized as follows. In Section II, we present the related work. Section III describes the decentralized constrained online learning problem and its performance metrics. We present COLDQ together with its doubly-bounded virtual queue and penalized Lyapunov drift designs in Section IV. We provide network regret and constraint violation analysis of COLDQ in Section V, followed by the discussion on its performance bounds in Section VI. Simulation results are provided in Section VII, followed by concluding remarks in Section VIII.

Supplementary materials: We provide the proofs of our Lemmas, Theorems, and Corollaries, together with the experimental parameter settings therein.

II. RELATED WORK

A. Centralized OCO with Fixed Constraints

Early OCO works used the projection operation to *strictly* satisfy the constraints at each time [8]-[17]. The seminal OCO work [8] achieved $\mathcal{O}(T^{\frac{1}{2}})$ static regret and a more meaningful $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret. The static regret was further improved to $\mathcal{O}(\log T)$ for strongly convex loss functions [9]. The dynamic regret has gained an increased attention in subsequent OCO works [10]-[14]. Some further improvements in the dynamic regret have been achieved by exploiting the strong convexity and smoothness properties [15]-[17].

To reduce the computational complexity incurred by the projection operation, the complicated short-term constraints were relaxed to *long-term* constraints, which need to be satisfied in the time-averaged manner [21]-[24]. The saddle-point-type algorithm proposed in [21] achieved $\mathcal{O}(T^{\frac{1}{2}})$ static regret and $\mathcal{O}(T^{\frac{3}{4}})$ constraint violation. Subsequent work [22] provided a trade-off between $\mathcal{O}(\max\{T^c, T^{1-c}\})$ static regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ constraint violation. For constraints satisfying the Slater's condition, which particularly excludes equality constraints, the virtual-queue-based algorithm in [23] achieved $\mathcal{O}(T^{\frac{1}{2}})$ static regret and the best-known $\mathcal{O}(1)$ constraint violation. Under periodic decision updates, the periodic queuing and gradient aggregation algorithm in [24] achieved $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret and $\mathcal{O}(T^{\frac{1}{2}})$ soft constraint violation. These works all adopted the *soft* constraint violation that allows *compensated* violations over time.

In contrast, aiming at limiting the *instantaneous* constraint violation, a stronger notion of *hard* constraint violation that does *not* allow any compensated violation over time was considered in [25]-[27]. The online algorithm proposed in [25] obtained $\mathcal{O}(\max\{T^c, T^{1-c}\})$ static regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ hard constraint violation. Improved $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret and $\mathcal{O}(T^{\frac{1+V_x}{4}})$ hard constraint violation were achieved in [26]. The best-known $\mathcal{O}(1)$ hard constraint violation and $\mathcal{O}(T^{\frac{1}{2}})$ static regret were achieved in [27], through a lower bounded virtual queue design. In this work, we consider OCO with time-varying constraints, which incorporates fixed constraints as a special case.

B. Centralized OCO with Time-Varying Constraints

Most prior works on centralized OCO with time-varying constraints bound the *soft* constraint violation [29]–[36]. For OCO problems with stochastic constraints, the virtual-queue-based algorithm proposed in [29] achieved $\mathcal{O}(T^{\frac{1}{2}})$ expected static regret and $\mathcal{O}(T^{\frac{1}{2}})$ expected soft constraint violation, under the Slater's condition. Similar $\mathcal{O}(T^{\frac{1}{2}})$ performance guarantees were obtained under a weaker assumption on the Lagrangian multiplier [30].

For *time-varying* constraints with unknown statistics, $\mathcal{O}(T^{\frac{1}{2}})$ static regret and $\mathcal{O}(T^{\frac{3}{4}})$ soft constraint violation were achieved in [32]. The modified saddle-point-type algorithm in [33] achieved $\mathcal{O}(\max\{T^{\frac{1+V_x}{2}}, T^{\frac{1+V_g}{2}}\})$ dynamic regret and $\mathcal{O}(\max\{T^{1-V_x}, T^{1-V_g}\})$ soft constraint violation, when the Slater's constant is sufficiently large. Another saddle-point-type algorithm in [34] achieved $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret and $\mathcal{O}(T^{\frac{3+V_x}{4}})$ soft constraint violation. The virtual-queue-based algorithm in [35] obtained $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret and $\mathcal{O}(\max\{T^{\frac{3}{4}}, T^{V_g}\})$ soft constraint violation without the Slater's condition. The delay-tolerant algorithm in [36] provided $\mathcal{O}(\max\{T^{\frac{1+V_x}{2}}, T^{V_g}\})$ dynamic regret and $\mathcal{O}(\max\{T^{\frac{1-V_x}{2}}, T^{V_g}\})$ soft constraint violation under the Slater's condition.

Some more recent works on constrained OCO bound the *hard* constraint violation [27], [28]. The rectified online optimization algorithm in [27] achieved $\mathcal{O}(T^{\frac{1}{2}+V_x})$ dynamic regret and $\mathcal{O}(T^{\frac{3}{4}})$ hard constraint violation. The best-known $\mathcal{O}(T^{\frac{1}{2}} \log T)$ hard constraint violation and $\mathcal{O}(T^{\frac{1}{2}})$ static regret were achieved in [28]. The constrained OCO algorithm in [28] was extended to the *optimistic* setting with predictions of the future information in [37]. Optimistic OCO is out of the scope of this work. We refer interested readers to [37] and references therein. *Unfortunately, as the dynamics of the loss and constraint functions decrease, i.e., $V_x \rightarrow 0$ and $V_g \rightarrow 0$, none of the above soft or hard constraint violation bounds for centralized OCO with time-varying constraints approaches to the best-known $\mathcal{O}(1)$ violation for fixed constraints.*

C. Decentralized Constrained OCO

With the increasing demand for large-scale machine learning, decentralized OCO has gained an increased attention. Early works aimed at achieving $\mathcal{O}(T^{\frac{1}{2}})$ static regret under dynamic networks [38]–[40]. Subsequent work [41] obtained $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret. For strongly convex and smooth loss functions, an improved $\mathcal{O}(T^{V_x})$ dynamic regret was achieved in [42] via primal-dual mirror descent. The above works all used projection-based operations to satisfy the constraints. Projection-free decentralized OCO algorithms were proposed in [43], [44]. However, these projection-free algorithms can only achieve $\mathcal{O}(T^{\frac{3}{4}})$ static regret.

Most existing works on decentralized *constrained* OCO bound the *soft* constraint violation [45]–[51]. Among them, decentralized OCO with *fixed* constraints was considered in [45]–[47]. For the special case that all the learners share a single fixed constraint function, the primal-dual algorithm in [45] provided a trade-off between $\mathcal{O}(T^{\frac{1}{2}+c})$ static regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ soft constraint violation. The decentralized

constrained OCO algorithm in [46] achieved $\mathcal{O}(T^{\frac{1}{2}+2c})$ static regret, and $\mathcal{O}(T^{1-\frac{c}{2}})$ soft constraint violation that allows the violations to be compensated among all the learners. The decentralized constrained OCO framework was extended to allow multiple constraints at each learner in [47]. The proposed online algorithm achieved $\mathcal{O}(T^{\frac{1}{2}})$ static regret and $\mathcal{O}(T^{\frac{3}{4}})$ soft constraint violation.

Decentralized OCO with *time-varying* constraints was studied in [48]–[51]. The primal-dual online algorithm in [48] achieved $\mathcal{O}(\max\{T^{c+V_x}, T^{1-c}\})$ dynamic regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ soft constraint violation. The soft constraint violation bound was improved to $\mathcal{O}(\max\{T^c, T^{1-c}\})$ under the Slater's condition. Another primal-dual online algorithm in [49] obtained $\mathcal{O}(\max\{T^{a+V_x}, T^{1-a+b}\})$ dynamic regret and $\mathcal{O}(\max\{T^{1-\frac{b}{2}}, T^{1-a+b}, T^{\frac{1+a-b+V_x}{2}}\})$ soft constraint violation. The online algorithm with bandit feedback in [50] achieved $\mathcal{O}(\max\{T^{c+V_x}, T^{1-c}\})$ dynamic regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ soft constraint violation. The online algorithm with gradient feedback in [51] obtained $\mathcal{O}(\max\{T^c, T^{1-c}\})$ static regret and $\mathcal{O}(\max\{T^{\frac{1+c}{2}}, T^{1-\frac{c}{2}}\})$ soft constraint violation.

Only a few works on decentralized constrained OCO bound the *hard* constraint violation [52]–[54]. The online algorithm in [52] achieved $\mathcal{O}(T^{\max\{c, 1-c\}})$ static regret for quadratic loss functions and $\mathcal{O}(T^{1-\frac{c}{2}})$ hard constraint violation for fixed constraints. Still for fixed constraints, $\mathcal{O}(T^{\frac{1}{2}})$ static regret and $\mathcal{O}(T^{\frac{3}{4}})$ hard constraint violation were achieved in [53]. For time-varying constraints, the same $\mathcal{O}(T^{\frac{1}{2}})$ static regret and $\mathcal{O}(T^{\frac{3}{4}})$ hard constraint violation were achieved in [54]. *Unfortunately still, as the system dynamics reduce, none of the above soft or hard constraint violation bounds for decentralized OCO with time-varying constraints approaches to $\mathcal{O}(1)$. Even for fixed constraints, it is open whether $\mathcal{O}(1)$ constraint violation can be achieved in the decentralized setting.*

Comparisons: In Tables I and II, we compare the performance bounds of COLDQ with the most relevant prior works. The comparison demonstrates that *COLDQ keeps pace with dynamics of both losses and constraints, under both centralized and decentralized settings.*

III. DECENTRALIZED CONSTRAINED ONLINE LEARNING

A. Problem Formulation

We consider decentralized constrained online learning over a networked system of N learners. The system operates in a time-slotted manner with time $t = 1, \dots, T$. At each time t , each learner n first selects a *local* decision $\mathbf{x}_t^n$ from a known feasible set $\mathcal{X} \subseteq \mathbb{R}^p$. The *local* loss function $f_t^n(\mathbf{x}) : \mathbb{R}^p \rightarrow \mathbb{R}$ and the *local* constraint function $\mathbf{g}_t^n(\mathbf{x}) = [g_t^{n,1}, \dots, g_t^{n,C}]^\top : \mathbb{R}^p \rightarrow \mathbb{R}^C$ are then revealed to the learner, incurring a local loss of $f_t^n(\mathbf{x}_t^n)$ and a local constraint violation of $\mathbf{g}_t^n(\mathbf{x}_t^n)$. Therefore, when learner n updates its local decision $\mathbf{x}_t^n$, it only has the *delayed* information of the previous local loss function $f_{t-1}^n(\mathbf{x})$ and the local constraint function $\mathbf{g}_{t-1}^n(\mathbf{x})$. Note that both the local loss function $f_t^n(\mathbf{x})$ and the local constraint function $\mathbf{g}_t^n(\mathbf{x})$ are unknown *a priori* and are allowed to change over time.

We are interested in decentralized online learning with a *global* loss function $f_t(\mathbf{x}) : \mathbb{R}^p \rightarrow \mathbb{R}$ at each time t . It is

TABLE I: Performance bounds for fixed constraints ($V_g = 0$).

Reference	Problem	Loss Function	Static Regret, Hard Constraint Violation	Dynamic Regret, Hard Constraint Violation
[26]	Centralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(T^{\frac{1}{4}})$	$\mathcal{O}(T^{\frac{1+V_x}{2}}), \mathcal{O}(T^{\frac{1+V_x}{4}})$
[27]	Centralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(1)$	$\mathcal{O}(T^{\frac{1}{2}+V_x})^*, \mathcal{O}(\log T)$
[53]	Decentralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(T^{\frac{3}{4}})$	Not given
Corollaries 3, 4	Decentralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(1)$	$\mathcal{O}(T^{\frac{1+V_x}{2}}), \mathcal{O}(1)$
[26]	Centralized	Strongly convex	$\mathcal{O}(\log T), \mathcal{O}(\log T)$	$\mathcal{O}(T^{\frac{1+V_x}{2}}), \mathcal{O}(T^{\frac{1+V_x}{4}})$
[27]	Centralized	Strongly convex	$\mathcal{O}(\log T), \mathcal{O}(1)$	$\mathcal{O}(T^{\frac{1}{2}+V_x})^*, \mathcal{O}(\log T)$
[53]	Decentralized	Strongly Convex	$\mathcal{O}(\log T), \mathcal{O}(T^{\frac{1}{2}}(\log T)^{\frac{1}{2}})$	Not given
Corollaries 3, 6	Decentralized	Strongly Convex	$\mathcal{O}(\log T), \mathcal{O}(1)$	$\mathcal{O}(T^{\frac{1+V_x}{2}}), \mathcal{O}(1)$

TABLE II: Performance bounds for time-varying constraints ($V_g > 0$).

Reference	Problem	Loss Function	Static Regret, Hard Constraint Violation	Dynamic Regret, Hard Constraint Violation
[27]	Centralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(T^{\frac{3}{4}})$	$\mathcal{O}(T^{\frac{1}{2}+V_x})^*, \mathcal{O}(T^{\frac{3}{4}})^{\dagger}$
[28]	Centralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(T^{\frac{1}{2}} \log T)$	Not given
[54]	Decentralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(T^{\frac{3}{4}})$	Not given
Corollaries 1, 2	Decentralized	Convex	$\mathcal{O}(T^{\frac{1}{2}}), \mathcal{O}(\min\{T^{\frac{3}{4}}, TV_g\})$	$\mathcal{O}(T^{\frac{1+V_x}{2}}), \mathcal{O}(\min\{T^{\frac{3+V_x}{4}}, TV_g\})$
[27]	Centralized	Strongly convex	$\mathcal{O}(\log T), \mathcal{O}(T^{\frac{1}{2}}(\log T)^{\frac{1}{2}})$	$\mathcal{O}(T^{\frac{1}{2}+V_x})^*, \mathcal{O}(T^{\frac{3}{4}})^{\ddagger}$
[28]	Centralized	Strongly convex	$\mathcal{O}(\log T), \mathcal{O}(T^{\frac{1}{2}}(\log T)^{\frac{1}{2}})$	Not given
[54]	Decentralized	Strongly Convex	$\mathcal{O}(T^c), \mathcal{O}(T^{1-\frac{\epsilon}{2}})$	Not given
Corollaries 1, 5	Decentralized	Strongly Convex	$\mathcal{O}(\log T), \mathcal{O}(\min\{T^{\frac{1}{2}}(\log T)^{\frac{1}{2}}, TV_g\})$	$\mathcal{O}(T^{\frac{1+V_x}{2}}), \mathcal{O}(\min\{T^{\frac{3+V_x}{4}}, TV_g\})$

* The $\mathcal{O}(T^{\frac{1}{2}+V_x})$ dynamic regret in [27] is achieved without the knowledge of V_x .

defined as the average of the local loss functions in the entire network, given by

$$f_t(\mathbf{x}) = \frac{1}{N} \sum_{n=1}^N f_t^n(\mathbf{x}). \quad (1)$$

The goal of the learners is to select online local decision sequences that minimize the *network-wise* accumulated global loss under *individual* time-varying constraints. This gives rise to the following decentralized time-varying constrained online learning problem

$$\begin{aligned} \mathbf{P}: \quad & \min_{\{\mathbf{x}_t^n \in \mathcal{X}\}} \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T f_t(\mathbf{x}_t^n) \\ \text{s.t.} \quad & \mathbf{g}_t^n(\mathbf{x}_t^n) \preceq \mathbf{0}, \quad \forall t, \forall n. \end{aligned} \quad (2)$$

When $\mathbf{g}_t^n(\mathbf{x}) = \mathbf{g}^n(\mathbf{x}), \forall t, \forall n$, $\mathbf{P}$ becomes the decentralized online learning problem with fixed constraints.

In decentralized online learning, it is impossible for each learner n to find the global minimizer of the global loss function $f_t(\mathbf{x})$ at each time t , without communicating with the other learners. We therefore consider the learners can communicate their local decisions $\{\mathbf{x}_t^n\}$ over the network. We model the communication network topology as a *time-varying directed* graph $\mathcal{G}_t = (\mathcal{N}, \mathcal{E}_t)$, where $\mathcal{N} = \{1, \dots, N\}$ is the set of learners and $\mathcal{E}_t \subseteq \mathcal{N} \times \mathcal{N}$ is the set of edges at time t . A directed edge (n, k) exists if learner n can communicate its local decision $\mathbf{x}_t^n$ to learner k at time t . Let $\mathcal{N}_{\text{in}}^n(\mathcal{G}_t) = \{k | (k, n) \in \mathcal{E}_t\}$ be the set of *in-neighbors* and $\mathcal{N}_{\text{out}}^n(\mathcal{G}_t) = \{k | (n, k) \in \mathcal{E}_t\}$ be the set of *out-neighbors* of learner n at time t .

For decentralized minimization of the global loss, each learner first updates a local decision $\mathbf{x}_t^n$ and then a local *estimator* $\mathbf{y}_t^n$ of the global minimizer at each time t . Each local estimator $\mathbf{y}_t^n$ is averaged over the local decisions $\{\mathbf{x}_t^k\}$ received from the in-neighbors $\mathcal{N}_{\text{in}}^n(\mathcal{G}_t)$ of learner n , given by

$$\mathbf{y}_t^n = \sum_{k=1}^N p_t^{n,k} \mathbf{x}_t^k, \quad (3)$$

where $p_t^{n,k} \geq 0$ is the *edge weight* from learner k to learner n that satisfies $p_t^{n,k} > 0$ if $(k, n) \in \mathcal{E}_t$ and $p_t^{n,k} = 0$ otherwise. When learner n updates its local decision $\mathbf{x}_t^n$, it only has the *delayed* information of the previous global estimator $\mathbf{y}_{t-1}^n$ and the previous edge weights $\{p_{t-1}^{n,k}\}$. Note that the edge weights $\{p_t^{n,k}\}$, representing the communication network topology, are also unknown *a priori* and are allowed to change over time.

B. Assumptions

We make some standard assumptions on the local loss functions $\{f_t^n(\mathbf{x})\}$, local constraint functions $\{\mathbf{g}_t^n(\mathbf{x})\}$, and feasible set $\mathcal{X}$ in the literature of OCO with hard constraints [26]-[28], [53], [54].

Assumption 1. (Bounded Set) The feasible set $\mathcal{X}$ is convex and bounded, i.e., $\exists R > 0$, such that

$$\|\mathbf{x} - \mathbf{y}\| \leq R, \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{X}. \quad (4)$$

Assumption 2. (Bounded Subgradient) The local loss functions $\{f_t^n(\mathbf{x})\}$ are convex with bounded subgradient over $\mathcal{X}$, i.e., $\exists D_1 > 0$, such that

$$\|\nabla f_t^n(\mathbf{x})\| \leq D_1, \quad \forall \mathbf{x} \in \mathcal{X}, \forall t, \forall n. \quad (5)$$

Assumption 3. The local constraint functions $\{g_t^n(\mathbf{x})\}$ are convex over $\mathcal{X}$ and satisfy the following:

3.1 (Bounded Value) $g_t^{n,c}(\mathbf{x})$ is bounded over $\mathcal{X}$, i.e., $\exists G > 0$, such that

$$|g_t^{n,c}(\mathbf{x})| \leq G, \quad \forall \mathbf{x} \in \mathcal{X}, \quad \forall t, \forall n, \forall c. \quad (6)$$

3.2 (Lipschitz Continuous) $g_t^{n,c}(\mathbf{x})$ is Lipschitz continuous over $\mathcal{X}$, i.e., $\exists D_2 > 0$, such that

$$|g_t^{n,c}(\mathbf{x}) - g_t^{n,c}(\mathbf{y})| \leq D_2 \|\mathbf{x} - \mathbf{y}\|, \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{X}, \quad \forall t, \forall n, \forall c. \quad (7)$$

Note that we do not require the commonly assumed Slater's condition [29], [33], [36] on each of the local constraint function at each time, i.e., $\exists \tilde{\mathbf{x}}_t \in \mathcal{X}$ and $\delta > 0$, such that $\mathbf{g}_t^n(\tilde{\mathbf{x}}_t) \prec -\delta \mathbf{1}$, $\forall t, \forall n$, or any of its relaxed version [30]. The Slater's condition, i.e., the existence of a shared interior point assumption, excludes equality constraints that are common in many practical applications.

In decentralized learning, the learners communicate their local decisions $\{\mathbf{x}_t^n\}$ according to the network topology graph $\mathcal{G}_t$, and assign edge weights $\{p_t^{n,k}\}$ to update their local estimators $\{\mathbf{y}_t^n\}$ of the global minimizer at each time t . To ensure sufficient communication among the learners, we make some standard assumptions on the network topology graphs and the edge weights [53]-[57].

Assumption 4. The network topology graphs $\{\mathcal{G}_t\}$ and the edge weights $\{p_t^{n,k}\}$ satisfy the following:

4.1 (Minimum Weight) The directed edges have a minimum edge weight, i.e., $\exists w \in (0, 1)$, such that

$$p_t^{n,k} \geq w, \quad \forall (k, n) \in \mathcal{E}_t, \quad \forall t. \quad (8)$$

4.2 (Doubly Stochastic) The edge weight matrix is doubly stochastic, i.e.,

$$\sum_{n=1}^N p_t^{n,k} = \sum_{k=1}^N p_t^{n,k} = 1, \quad \forall t, \forall n, k. \quad (9)$$

4.3 (Connectivity) The directed graph $(\mathcal{N}, \bigcup_{i=0}^{B-1} \mathcal{E}_{t+i})$ is strongly connected, i.e., $\exists B \in \mathbb{Z} > 1$, such that

$$(n, k) \in \mathcal{E}_t \cup \mathcal{E}_{t+1} \cup \dots \cup \mathcal{E}_{t+B-1}, \quad \forall (n, k), \forall t. \quad (10)$$

We can adopt some existing methods to ensure the edge weight matrix is doubly stochastic [58]. We make the common connectivity assumption to ensure each learner n can receive a local decision made by any other learner k at least once in every B consecutive time slots.

C. Performance Metrics

Finding an optimal solution to $\mathbf{P}$ is impossible since the current information about the global loss function $f_t(\mathbf{x})$ and the local constraint function $\mathbf{g}_t^n(\mathbf{x})$ is not available when each learner n selecting its local decision $\mathbf{x}_t^n$ at each time t . Instead, the OCO literature measures the performance of a decentralized constrained online algorithm, by comparing it with some solution benchmarks. There are two commonly used benchmarks.

The first one is the *fixed offline* global solution benchmark¹

$$\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{X}} \left\{ \sum_{t=1}^T f_t(\mathbf{x}) \mid \mathbf{g}_t^n(\mathbf{x}) \preceq \mathbf{0}, \forall t, \forall n \right\}, \quad (11)$$

where $\mathbf{x}^*$ is computed assuming all the information of the global loss functions $\{f_t(\mathbf{x})\}$ and the local constraint functions $\{\mathbf{g}_t^n(\mathbf{x})\}$ is known a posteriori over the entire T time slots. The resulting network *static* regret, which measures the performance gap between the online local decision sequences $\{\mathbf{x}_t^n\}$ and the fixed offline global solution benchmark $\mathbf{x}^*$, is defined as

$$\text{REG}_s(T) = \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T f_t(\mathbf{x}_t^n) - \sum_{t=1}^T f_t(\mathbf{x}^*). \quad (12)$$

The static regret (12) was adopted in [28] for centralized OCO with hard constraints, while [53], [54] used a special case of it when all the learners share the same constraint function, for time-invariant and time-varying constraints, respectively. However, in dynamic environments, the static regret may not serve as a strong indicator of the actual performance of an online algorithm.

The second one is the *dynamic online* global solution benchmark

$$\mathbf{x}_t^* \in \min_{\mathbf{x} \in \mathcal{X}} \{f_t(\mathbf{x}) \mid \mathbf{g}_t^n(\mathbf{x}) \preceq \mathbf{0}, \forall n\}, \quad (13)$$

where $\mathbf{x}_t^*$ is computed using current information of the global loss function $f_t(\mathbf{x})$ and the local constraint functions $\{\mathbf{g}_t^n(\mathbf{x})\}$ at each time t . The resulting network *dynamic* regret, which measures the performance gap between $\{\mathbf{x}_t^n\}$ and $\{\mathbf{x}_t^*\}$, is defined as

$$\text{REG}_d(T) = \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T f_t(\mathbf{x}_t^n) - \sum_{t=1}^T f_t(\mathbf{x}_t^*). \quad (14)$$

The dynamic regret (14) was adopted in [26], [27] for centralized OCO with fixed and time-varying hard constraints, respectively. Note that the difference between the network dynamic regret in (12) and the network static regret in (14) can scale linearly with T , i.e., $\text{REG}_d(T) - \text{REG}_s(T) = \mathcal{O}(T)$ [60]. For a thorough analysis, in this work we provide upper bounds on both the network dynamic regret and static regret.

Another one is *any feasible* solution benchmark $\mathbf{u}_t \in \{\mathbf{g}_t^n(\mathbf{x}) \preceq \mathbf{0}, \forall \mathbf{x} \in \mathcal{X}, \forall n\}$, which incorporates the above fixed offline solution benchmark $\mathbf{x}^*$ in (11) and dynamic online solution benchmark $\{\mathbf{x}_t^*\}$ in (13) as two special cases. The resulting *universal* regret has been studied for centralized OCO [12], [14], [17]. Our performance analysis in Section V can be readily extended to the network universal regret $\frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T f_t(\mathbf{x}_t^n) - \sum_{t=1}^T f_t(\mathbf{u}_t)$ by replacing $\mathbf{x}_t^*$ with $\mathbf{u}_t$.

¹It is known that *no* causal algorithm can achieve sublinear static regret to the benchmark $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{X}} \{\sum_{t=1}^T f_t(\mathbf{x}) \mid \sum_{t=1}^T \mathbf{g}_t^n(\mathbf{x}) \preceq \mathbf{0}\}$, which satisfies the *long-term* constraint even in the *centralized* setting [59]. It is possible to achieve sublinear static regret to a K -benchmark $\mathbf{x}_K^* \in \arg \min_{\mathbf{x} \in \mathcal{X}} \{\sum_{t=1}^T f_t(\mathbf{x}) \mid \sum_{\tau=t}^{t+K-1} \mathbf{g}_\tau^n(\mathbf{x}) \preceq \mathbf{0}, \forall 1 \leq t \leq T - K + 1\}$ and sublinear *soft* constraint violation, when $K = o(T)$ [31].

There are also two commonly used performance metrics to quantify how much the time-varying constraints (2) are violated. The first one is the *soft* constraint violation defined as

$$\text{VIO}_s(T) = \frac{1}{N} \sum_{n=1}^N \left\| \left[\sum_{t=1}^T \mathbf{g}_t^n(\mathbf{x}_t^n) \right]_+ \right\|, \quad (15)$$

where $[\cdot]_+$ is the element-wise projector onto the non-negative space. The soft constraint violation (15) allows the violation of constraints at individual time slots to be compensated over time [33], [45], [49], while [46], [50], [51] even allow compensated constraint violations among all the learners.

The second one is the *hard* constraint violation defined as

$$\text{VIO}_h(T) = \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T \left\| [\mathbf{g}_t^n(\mathbf{x}_t^n)]_+ \right\|. \quad (16)$$

This hard constraint violation does *not* allow the violation of a constraint at one time or one learner to be compensated by any other time [26]-[28], or any other learners [53], [54]. From the definitions of the soft and hard constraint violations in (15) and (16), we readily have $\text{VIO}_s(T) \leq \text{VIO}_h(T)$. In this work, we provide upper bounds on the hard constraint violation, which directly apply to the soft constraint violation as well.

D. Variation Measures

We aim to design a decentralized constrained OCO algorithm that simultaneously achieves sublinear network static regret $\text{REG}_s(T) = o(T)$, sublinear network dynamic regret $\text{REG}_d(T) = o(T)$, and sublinear hard constraint violation $\text{VIO}_h(T) = o(T)$. This implies that the local decision sequences yielded by our algorithm are asymptotically no worse than either the fixed offline or the dynamic online global solution benchmarks, while satisfying individual constraints in the time-averaged manner. This triple objective, however, can be intractable due to arbitrary variations of the global loss functions $\{f_t(\mathbf{x})\}$ and the local constraint functions $\{\mathbf{g}_t^n(\mathbf{x})\}$. It is therefore necessary to quantify the variations of the underlying decentralized constrained online learning problem **P**.

There are two common variation measures in the constrained OCO literature. The first one measures the variations of the dynamic online global solution benchmark $\{\mathbf{x}_t^*\}$ as

$$\sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| = \mathcal{O}(T^{V_x}), \quad (17)$$

where $V_x \in [0, 1]$ represents the time variability of the dynamic online global solution benchmark. It is commonly referred to as the *path length* in centralized constrained OCO [26], [33]-[36].

The second one measures the variations of the local constraint functions $\{\mathbf{g}_t^n(\mathbf{x})\}$ as

$$\sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| = \mathcal{O}(T^{V_g}), \quad (18)$$

where $V_g \in [0, 1]$ represents the time variability of the local constraint functions. It extends the variation measure on the

constraint functions for centralized constrained OCO [33], [35], [36] to the decentralized case. Note that for the fixed offline global solution benchmark, *i.e.*, $\mathbf{x}_t^* = \mathbf{x}^*, \forall t$, we have $V_x = 0$. Similarly, for fixed local constraint functions, *i.e.*, $\mathbf{g}_t^n(\mathbf{x}) = \mathbf{g}^n(\mathbf{x}), \forall t, \forall n$, we have $V_g = 0$.

IV. CONSTRAINED ONLINE LEARNING WITH DOUBLY-BOUNDED QUEUE (COLDQ)

We present the details of COLDQ for solving **P**. In COLDQ, we introduce a novel doubly-bounded virtual queue and a new penalized Lyapunov drift design, which will be shown to provide improved regret and constraint violation bounds for OCO with hard constraints.

A. Doubly-Bounded Virtual Queue

We introduce a local virtual queue vector $\mathbf{Q}_t^n = [Q_t^{n,1}, \dots, Q_t^{n,C}]^\top \in \mathbb{R}^C$, initialized as $\mathbf{Q}_1^n = \gamma_1 \mathbf{1}$, to track the amounts of local constraint violations for each learner n . At the end of each time $t > 1$, after observing the local constraint function $\mathbf{g}_t^n(\mathbf{x})$, each learner n updates $\mathbf{Q}_t^n$ as:

$$\mathbf{Q}_t^n = \max \{ (1 - \eta_t) \mathbf{Q}_{t-1}^n + [\mathbf{g}_t^n(\mathbf{x}_t^n)]_+, \gamma_t \mathbf{1} \}, \quad (19)$$

where $\eta_t \in (0, 1)$ is a non-increasing *penalty* factor on the previous virtual queue $\mathbf{Q}_{t-1}^n$, $\gamma_t \in (0, \frac{G}{\eta_t})$ is a non-decreasing *minimum* length of the current virtual queue $\mathbf{Q}_t^n$, and $\max\{\mathbf{a}, \mathbf{b}\}$ is applied pairwise. Note that for the local virtual queue $\mathbf{Q}_t^n$ update at the *end* of time t , the current local constraint function $\mathbf{g}_t^n(\mathbf{x})$ information is readily available at each learner n . This is in contrast to the local decision $\mathbf{x}_t^n$ update at the *beginning* of time t , where only the delayed local constraint function $\mathbf{g}_{t-1}^n(\mathbf{x})$ information is available.

Our virtual queue updating rule (19) includes an additional penalty term $\eta_t \mathbf{Q}_{t-1}^n$ on the previous virtual queue to avoid the current virtual queue $\mathbf{Q}_t^n$ from becoming excessively large. In the following lemma, we show that, even *without* the commonly assumed Slater's condition – the actual Slater constant can be 0 – the penalty factor η_t can serve as a *virtual* Slater constant in the virtual queue upper bound. Furthermore, the minimum virtual queue length γ_t in (19) ensures that the virtual queue length is always strictly positive, which prevents the constraint violation from being overly large.

Lemma 1. (Bounds on Virtual Queue) Under Assumption 3.1, the virtual queue updated by each learner n via (19) has both a lower bound and an upper bound for each time t , given by

$$\gamma_t \mathbf{1} \preceq \mathbf{Q}_t^n \preceq \frac{G}{\eta_t} \mathbf{1}. \quad (20)$$

Proof: See Appendix A in the supplementary materials. ■

Lemma 1 shows that, without requiring the Slater's condition, our virtual queue updating rule (19) leads to both a *lower* bound and an *upper* bound on the virtual queue. These virtual queue bounds, however, cannot be directly translated into a bound on the hard constraint violation $\text{VIO}_h(T)$ in (16). In the following section, we will establish a connection between our virtual queue $\mathbf{Q}_t^n$ and the hard constraint violation $\text{VIO}_h(T)$,

via a new penalized Lyapunov drift approach that leverages both the lower and upper bounds of the virtual queue in (20).

Existing constrained OCO works that adopt the virtual queue technique can be divided into two groups: Upper bounded virtual queues were introduced to bound the soft and hard constraint violations in [23] [29]-[31], [35], [36], [48]-[50] and [26], [54], respectively; A lower bounded virtual queue was introduced to bound the hard constraint violation in [27]. The constraint violation was bounded either indirectly via the standard Lyapunov drift analysis that uses the virtual queue upper bound only [23], [29]-[31], [35], [36], or directly via the virtual queue lower [27] or upper bound [26], [48]-[50], [54]. In contrast, our virtual queue yields *both* lower and upper bounds. Together with a new penalized Lyapunov drift analysis that leverages both bounds, COLDQ provides improved violation bounds over the current best results.

B. Penalized Lyapunov Drift

Based on (19), we define a new *penalized* Lyapunov drift for each learner n and time $t > 1$ as

$$\Delta_{t-1}^n = \frac{1}{2} \|\mathbf{Q}_t^n - \gamma_t \mathbf{1}\|^2 - \frac{1}{2} \|\mathbf{Q}_{t-1}^n - \gamma_{t-1} \mathbf{1}\|^2. \quad (21)$$

Compared with the standard Lyapunov drift that uses the quadratic virtual queue as the Lyapunov function, each virtual queue $\mathbf{Q}_t^n$ is penalized by its lower bound γ_t in (21). The following lemma establishes an upper bound for Δ_{t-1}^n , leveraging both the lower and upper bounds of $\mathbf{Q}_t^n$ in (20).

Lemma 2. (Bound on Penalized Lyapunov Drift) Under Assumption 3.1, the penalized Lyapunov drift (21) at each learner n is upper bounded for any $t > 1$ by

$$\begin{aligned} \Delta_{t-1}^n &\leq (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ - \gamma_{t-1} \mathbf{1}^\top [\mathbf{g}_t^n(\mathbf{x}_t^n)]_+ \\ &\quad + \frac{G\sqrt{C}}{\eta_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| + 2G^2C, \end{aligned} \quad (22)$$

where C is the number of local constraints.

Proof: See Appendix B in the supplementary materials. ■

The penalized Lyapunov drift upper bound in (22) comprises two key terms. The second term on the right-hand side (RHS) of (22) accounts for the hard constraint violation $[\mathbf{g}_t^n(\mathbf{x}_t^n)]_+$, scaled by the virtual queue lower bound γ_{t-1} . The third term on the RHS of (22) captures the variation of two consecutive constraint functions $\|\mathbf{g}_t^n(\mathbf{x}_t^n) - \mathbf{g}_{t-1}^n(\mathbf{x}_t^n)\|$, scaled by the virtual queue upper bound $\frac{G}{\eta_{t-1}}$. These two terms are critical for relating the hard constraint violation $\text{VIO}_h(T)$ to the constraint variation measure V_g in (18).

C. The COLDQ Algorithm

We convert the original decentralized constrained online learning problem $\mathbf{P}$ into solving a *per-slot local* optimization problem $\mathbf{P}_t^n$, to determine the local decision $\mathbf{x}_t^n$ for each learner n at each time $t > 1$, given by

$$\begin{aligned} \mathbf{P}_t^n : \min_{\mathbf{x} \in \mathcal{X}} & \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x} - \mathbf{y}_{t-1}^n \rangle + \alpha_{t-1} \|\mathbf{x} - \mathbf{y}_{t-1}^n\|^2 \\ & + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x})]_+, \end{aligned}$$

where $\langle \mathbf{a}, \mathbf{b} \rangle$ denotes the inner product operation, and α_{t-1} is a non-decreasing *step-size* parameter, and $\mathbf{y}_{t-1}^n = \sum_{k=1}^N p_{t-1}^{n,k} \mathbf{x}_{t-1}^k$ defined in (3) is the previous local estimator of the global minimizer. Note that we use $\mathbf{y}_{t-1}^n$ instead of $\mathbf{x}_{t-1}^n$ in $\mathbf{P}_t^n$, since $\mathbf{y}_{t-1}^n$ is an aggregation of the local decisions received from the in-neighbors $\mathcal{N}_{\text{in}}^n(\mathcal{G}_{t-1})$ of learner n , and thus estimates the global minimizer better than $\mathbf{x}_{t-1}^n$. However, we do not use the gradient of $\mathbf{y}_{t-1}^n$ in $\mathbf{P}_t^n$, since the calculation of $\nabla f_{t-1}^n(\mathbf{y}_{t-1}^n)$ may not be completed by the end of time $t-1$.² This is because local decision aggregation happens only after local decision update, and both the gradient calculation and decision aggregation can be time-consuming.

Minimizing $\langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x} - \mathbf{y}_{t-1}^n \rangle + \alpha_{t-1} \|\mathbf{x} - \mathbf{y}_{t-1}^n\|^2$ over $\mathcal{X}$ in $\mathbf{P}_t^n$ is equivalent to performing projected local gradient descent $\mathcal{P}_{\mathcal{X}}\{\mathbf{y}_{t-1}^n - \frac{1}{2\alpha_{t-1}} \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)\}$, where $\mathcal{P}_{\mathcal{X}}\{\mathbf{a}\} = \arg \min_{\mathbf{b} \in \mathcal{X}} \|\mathbf{a} - \mathbf{b}\|^2$ is the projector onto set $\mathcal{X}$. The optimal solution to $\mathbf{P}_t^n$ depends on the amount of constraint violation induced by this projected gradient descent in the following two cases.

• If $\mathbf{g}_{t-1}^n(\mathcal{P}_{\mathcal{X}}\{\mathbf{y}_{t-1}^n - \frac{1}{2\alpha_{t-1}} \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)\}) \preceq \mathbf{0}$, i.e., the projected gradient descent does not incur any constraint violation, the optimal solution to $\mathbf{P}_t^n$ is

$$\mathbf{x}_t^n = \mathcal{P}_{\mathcal{X}}\left\{\mathbf{y}_{t-1}^n - \frac{1}{2\alpha_{t-1}} \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)\right\}. \quad (23)$$

• If $\mathbf{g}_{t-1}^n(\mathcal{P}_{\mathcal{X}}\{\mathbf{y}_{t-1}^n - \frac{1}{2\alpha_{t-1}} \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)\}) \not\preceq \mathbf{0}$, i.e., at least one constraint is violated, the gradient descent direction is shifted towards minimizing the term $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x})]_+$ in $\mathbf{P}_t^n$ to reduce the constraint violation. The virtual queue $\mathbf{Q}_{t-1}^n$ balances between minimizing the loss function and reducing the constraint violation. The exact form of the optimal solution depends on the constraint function $\mathbf{g}_{t-1}^n(\mathbf{x})$. Later in Section V, the non-negativity of $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x})]_+$ and Lemma 2 allow us to *separately* bound the network dynamic regret in Theorem 1 and the hard constraint violation in Theorem 3 via Lemma 4.

From the penalized Lyapunov drift upper bound established in (22), we can see the intuition behind solving $\mathbf{P}_t^n$. Specifically, the objective is to greedily minimize the upper bound on the following *penalized drift plus penalty* term:

$$\underbrace{\Delta_{t-1}^n}_{\text{penalized drift}} + \underbrace{\langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x} - \mathbf{y}_{t-1}^n \rangle + \alpha_{t-1} \|\mathbf{x} - \mathbf{y}_{t-1}^n\|^2}_{\text{penalty}}.$$

Note that the last two terms on the RHS of the penalized Lyapunov drift upper bound in (22) are independent of $\mathbf{x}_t^n$, and the second term is omitted since $\mathbf{g}_t^n(\mathbf{x})$ is not available when choosing $\mathbf{x}_t^n$. This is similar to the Lyapunov optimization approach [61] that minimizes a drift plus penalty term at each time. Different from the standard Lyapunov optimization, we additionally penalize the virtual queue lower bound in the Lyapunov drift (21), instead of using the quadratic virtual queue as the Lyapunov function. Furthermore, we design the penalty term to enable local gradient descent from the

²Using $\nabla f_{t-1}^n(\mathbf{y}_{t-1}^n)$ instead of $\nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)$ in $\mathbf{P}_t^n$ yields the same $\mathcal{O}(T^{\frac{1+V_g}{2}})$ dynamic regret, which also matches the $\Omega(T^{\frac{1+V_g}{2}})$ dynamic regret lower bound established in [12] for centralized OCO.

Algorithm 1 Constrained Online Learning with Doubly-bounded Queue (COLDQ)

Input: Non-decreasing step size $\{\alpha_t > 0\}$, non-increasing penalty factor $\{0 < \eta_t < 1\}$, and non-decreasing minimum virtual queue length $\{0 < \gamma_t \leq \frac{C}{\eta_t}\}$.

Output: Local decision sequences $\{\mathbf{x}_t^n\}$.

```

1: Initialize  $\mathbf{x}_1^n \in \mathcal{X}$  with  $\|\mathbf{x}_1^n\| \leq R$ , and  $\mathbf{Q}_1^n = \gamma_1 \mathbf{1}, \forall n$ .
2: for each time  $t = 2, \dots, T$  do
3:   for each  $n = 1, \dots, N$  in parallel do
4:     Update  $\mathbf{x}_t^n$  by solving  $\mathbf{P}_t^n$ .  $\triangleright$  Decision update
5:     Observe  $\nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)$  and  $\mathbf{g}_t^n(\mathbf{x}_t^n)$ .
6:     Update  $\mathbf{Q}_t^n$  via (19).  $\triangleright$  Queue update
7:     Send  $\mathbf{x}_t^n$  to  $k \in \mathcal{N}_{\text{out}}^n(\mathcal{G}_t)$ .  $\triangleright$  Decision transmission
8:     Receive  $\mathbf{x}_t^k$  from  $k \in \mathcal{N}_{\text{in}}^n(\mathcal{G}_t)$  and assign  $p_{t-1}^{n,k}$ .
9:     Update  $\mathbf{y}_t^n$  via (3).  $\triangleright$  Decision consensus
10:   end for
11: end for

```

estimator of the global minimizer for global loss minimization, instead of using the loss function itself as the penalty term.

In Algorithm 1, we give the pseudocode of COLDQ, which consists of four main steps.

- *Decision update:* At line 4, each learner n updates its local decision $\mathbf{x}_t^n$ at the beginning of each time t , based only on the gradient of the previous local loss function $\nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)$ and the previous local constraint function $\mathbf{g}_{t-1}^n(\mathbf{x})$. This decision update is designed to balance the accumulated global loss minimization and the local constraint violation control.
- *Queue update:* At line 6, each learner n updates its local virtual queue $\mathbf{Q}_t^n$ at the end of each time t , after observing the current constraint function $\mathbf{g}_t^n(\mathbf{x}_t^n)$. This queue update is to track the amount of local hard constraint violations.
- *Decision transmission:* At line 7, each learner n sends its updated local decision $\mathbf{x}_t^n$ to each of its out-neighbors $k \in \mathcal{N}_{\text{out}}^n(\mathcal{G}_t)$ at the end of time t . This decision transmission incurs a total communication overhead of $p \sum_{n \in \mathcal{N}} |\mathcal{N}_{\text{out}}^n(\mathcal{G}_t)|$, where p is the decision dimension.³
- *Decision consensus:* At line 9, each learner n updates the local estimator $\mathbf{y}_t^n$ of the global minimizer at the end of time t , by averaging the local decisions $\{\mathbf{x}_t^k\}$ received from its in-neighbors $\mathcal{N}_{\text{in}}^n(\mathcal{G}_t)$ over the assigned edge weights $\{p_{t-1}^{n,k}\}$. This decision consensus does not require the possibly private loss and constraint information from the other learners.

V. PERFORMANCE BOUNDS OF COLDQ

In this section, we provide general upper bounds on the static regret (12), dynamic regret (14), and hard constraint violation (16) yielded by COLDQ for both convex and strongly-convex loss functions. We will discuss the algorithm parameters $\alpha_t, \eta_t, \gamma_t$ to derive the best performance bounds for COLDQ in Section VI.

³For problems with a large p , we can further design constraint functions $\{g_t^n(\mathbf{x})\}$ to limit the communication resources [47], [55], [56]. As an example, in Section VII-B, we adopt conditional entropy as the constraint function to limit the number of transmitted bits for quantized digital communication.

A. Preliminary Analysis

In the following lemma, we provide an upper bound on the averaged optimal objective values of the per-slot local problems $\{\mathbf{P}_t^n\}$, resulting from the local decisions $\{\mathbf{x}_t^n\}$ generated by COLDQ.

Lemma 3. (Bound on $\mathbf{P}_t^n$ Objective) Under Assumptions 1-4, the online local decision sequence $\{\mathbf{x}_t^n\}$ generated by COLDQ satisfies the following inequality for any $t > 1$:

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N [\langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_t^n - \mathbf{y}_{t-1}^n \rangle + \alpha_{t-1} \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 \\
& \quad + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+] \\
& \leq \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n \rangle + 2R\alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \\
& \quad + \frac{\alpha_{t-1}}{N} \sum_{n=1}^N [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2]. \tag{24}
\end{aligned}$$

Proof: See Appendix C in the supplementary materials. ■

In Lemma 3, we intentionally introduce the dynamic online global solution benchmark $\mathbf{x}_{t-1}^*$ in the first term on the RHS of (24). This is to relate to the per-slot global loss difference $f_{t-1}(\mathbf{x}_{t-1}^n) - f_{t-1}(\mathbf{x}_{t-1}^*)$ between the online local decision $\mathbf{x}_{t-1}^n$ and $\mathbf{x}_{t-1}^*$ in the following lemma. The second term on the RHS of (24) establishes a connection to the variation measure of the dynamic online global solution benchmark (17). The last term on the RHS of (24) is telescoping over time. Using the results in Lemma 3, the following lemma establishes a per-slot performance guarantee of the COLDQ algorithm.

Lemma 4. (Bound on Per-Slot Performance) Under Assumptions 1-4, the online local decision sequences $\{\mathbf{x}_t^n\}$ generated by COLDQ satisfy the following inequality for any $t > 1$:

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N [f_{t-1}(\mathbf{x}_{t-1}^n) - f_{t-1}(\mathbf{x}_{t-1}^*) + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+] \\
& \leq \frac{1}{N} \sum_{n=1}^N [4D_1 \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| + D_1 \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\| \\
& \quad - \alpha_{t-1} \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2] + 2R\alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \\
& \quad + \frac{\alpha_{t-1}}{N} \sum_{n=1}^N [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2], \tag{25}
\end{aligned}$$

where $\bar{\mathbf{x}}_{t-1} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_{t-1}^n$.

Proof: See Appendix D in the supplementary materials. ■

Note that $\bar{\mathbf{x}}_{t-1}$ is the *virtual* global decision averaged over the entire network $\bar{\mathbf{x}}_{t-1}$. The difference between the actual local decision and the virtual global decision $\|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\|$ in the first term on the RHS of (25), can be transformed into the difference between the local decision and the local estimator $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|$ using Lemma 7 in Appendix E of the supplementary materials. As a result, the first term on the RHS of (25) is in a single form of $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|$, and the constants before it can then be canceled out by properly setting α_{t-1} .

Lemma 4 is the key to bridge the per-slot local optimization problems $\{\mathbf{P}_t^n\}$ and the performance bounds of COLDQ. Using the results in Lemma 4, we can *separately* bound the network dynamic regret and hard constraint violation, by substituting two different lower bounds on $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+$ into the left-hand side (LHS) of (25). Note that adopting the soft constraint violation measure necessitates *jointly* bounding the regret and constraint violation.

B. Bounding Network Dynamic Regret and Static Regret

The virtual queue length is always strictly positive due to its lower bound in (20), and the hard constraint violation is non-negative by definition. Therefore, their product $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+$ in (25) of Lemma 4 is guaranteed to be non-negative. Unlike the analysis for soft constrained OCO algorithms, this unique property enables us to bound the dynamic regret of the COLDQ, without needing to explicitly consider the hard constraint violation. Using the results in Lemma 4, the following theorem provides an upper bound on the dynamic regret (14) for COLDQ, in terms of the variation measure of the dynamic online global solution benchmark $\sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| = \mathcal{O}(T^{V_x})$ in (17). As will be shown later in Section VI, the resulting network dynamic regret of COLDQ is $\mathcal{O}(T^{\frac{1+V_x}{2}})$ for any $V_x > 0$.

Theorem 1. (Network Dynamic Regret Bound) Under Assumptions 1-4, the network dynamic regret of COLDQ is upper bounded by

$$\text{REG}_d(T) \leq 2R \sum_{t=2}^T \alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{\varepsilon_1^2}{4} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} + \alpha_{T-1} R^2 + \varepsilon_1 R, \quad (26)$$

where $\varepsilon_1 = D_1(\frac{4\rho N}{1-\lambda} + 9)$ with $\rho = \frac{2(1+w^{-B_0})}{1-w^{B_0}} \geq 2$, $\lambda = (1-w^{B_0})^{\frac{1}{B_0}} \in (0, 1)$, and $B_0 = (N-1)B$ are some constants introduced from Lemma 6 in Appendix E of the supplementary materials on the edge weight transition matrix.

Proof: See Appendix F in the supplementary materials. ■

In the centralized setting, the dynamic regret of COLDQ can be bounded directly through the per-slot optimization problem to build a connection between the online decision and the dynamic online benchmark [1]. In the decentralized setting, we connect $\mathbf{x}_t^n$ and $\mathbf{x}_t^*$ by taking two additional steps to bound $\|\mathbf{x}_t^n - \bar{\mathbf{x}}_t\|$ through $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2$ and then properly choosing ε_1 to cancel the constants before $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2$ [49], [54]. Note that when we bound the dynamic regret, we do not need to worry about the hard constraint violation bound, which is different from [49], [54].

Using the results in Theorem 1, the following theorem provides an upper bound on the network static regret for COLDQ.

Theorem 2. (Network Static Regret Bound) Under Assumptions 1-4, the static regret of COLDQ is upper bounded by

$$\text{REG}_s(T) \leq \frac{\varepsilon_1^2}{4} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} + \alpha_{T-1} R^2 + \varepsilon_1 R. \quad (27)$$

Proof: We conducted a unified network regret analysis that is applicable to both the dynamic online and the static offline global solution benchmarks in Theorem 1. Replacing all $\{\mathbf{x}_t^*\}$ with $\mathbf{x}^*$ in the proof of Theorem 1, we can show that (27) readily holds. ■

As will be shown later in Section VI, the resulting network static regret of COLDQ for convex loss functions is $\mathcal{O}(T^{\frac{1}{2}})$.

C. Bounding Hard Constraint Violation with V_g

The following theorem establishes a bound on the hard constraint violation (16) of COLDQ, using the variation measure of the local constraint function $\sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| = \mathcal{O}(T^{V_g})$ in (18). This is achieved by converting the term $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+$ in (25) of Lemma 4 to $[\mathbf{g}_t^n(\mathbf{x}_t^n)]_+$ through the penalized Lyapunov drift upper bound (22) in Lemma 2, which introduces the per-slot variation of the local constraint function $\max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\|$. As will be shown later in Section VI, the resulting hard constraint violation is $\mathcal{O}(T^{V_g})$ if $V_g > 0$, *i.e.*, when bounding time-varying constraints, and is $\mathcal{O}(1)$ if $V_g = 0$, *i.e.*, when bounding fixed constraints.

Theorem 3. (Hard Constraint Violation Bound #1) Under Assumptions 1-4, the hard constraint violation of COLDQ is upper bounded by

$$\text{VIO}_h(T) \leq G\sqrt{C} \sum_{t=2}^T \frac{1}{\eta_{t-1}\gamma_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| + 3R^2 \sum_{t=2}^T \frac{\alpha_{t-1}}{\gamma_{t-1}} + (6RD_1 + 2G^2C) \sum_{t=2}^T \frac{1}{\gamma_{t-1}} + G\sqrt{C}. \quad (28)$$

Proof: See Appendix G in the supplementary materials. ■

Extending the hard constraint violation bound in the centralized setting [1] to the decentralized setting is straightforward since the constraints are on the local decisions and the impact of the additional terms $\|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\|$ and $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|$ can be eliminated by the virtual queue lower bound. Note that different from [27], we do not need to worry about the dynamic regret bound when we bound the hard constraint violation.

D. Bounding Hard Constraint Violation with V_x

The following theorem establishes another bound on the hard constraint violation (16) of COLDQ, using the variation measure of the dynamic online global solution benchmark $\sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| = \mathcal{O}(T^{V_x})$ in (17). This is achieved by first converting the bound on the hard constraint violation $\|[\mathbf{g}_t^n(\mathbf{x}_t^n)]_+\|$ into a bound on the difference between the current local decision and the previous local estimator $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|$. The results of Lemmas 3 and 4 are then used to bound $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|$. This process relates the per-slot variation of the dynamic online global solution benchmark $\|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\|$ to the hard constraint violation bound. As will be shown later in Section VI, the resulting hard constraint violation is $\mathcal{O}(T^{\frac{3+V_x}{4}})$ if $V_x > 0$, *i.e.*, when bounding the dynamic regret, and is $\mathcal{O}(T^{\frac{3}{4}})$ if $V_x = 0$, *i.e.*, when bounding the static regret.

Theorem 4. (Hard Constraint Violation Bound #2) Under Assumptions 1-4, for any $\beta > 0$, the hard constraint violation of COLDQ is also upper bounded by

$$\begin{aligned} \text{VIO}_h(T) &\leq 2\varepsilon_2 R \sum_{t=2}^T \beta \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \varepsilon_2 R D_1 \sum_{t=2}^T \frac{\beta}{\alpha_{t-1}} \\ &\quad + R^2 \sum_{t=2}^T \frac{\alpha_{t-1}}{\gamma_{t-1}} + R D_1 \sum_{t=2}^T \frac{1}{\gamma_{t-1}} \\ &\quad + \varepsilon_2 \frac{T}{\beta} + \varepsilon_2 R^2 \beta + 2\varepsilon_2 R + G\sqrt{C}, \end{aligned} \quad (29)$$

where $\varepsilon_2 = D_2[C + \sqrt{C}(\frac{\rho N}{1-\lambda} + 2)]$ is a constant.

Proof: See Appendix H in the supplementary materials. ■

The hard constraint violation can be bounded through the per-slot optimization problem to bound $\|\mathbf{x}_t - \mathbf{x}_{t-1}\|$ only for centralized constrained OCO [27]. In the decentralized setting, the necessity of having the local estimator $\mathbf{y}_t^n$ in the per-slot local optimization problem $\mathbf{P}_t^n$ to minimize the global loss breaks the direct connection between $\mathbf{x}_t$ and $\mathbf{x}_{t-1}$. Therefore, we first bound $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|$ through the bound on $\mathbf{P}_t^n$ objective in Lemma 3, and then bound $\|\mathbf{x}_{t-1}^n - \mathbf{y}_{t-1}^n\|$ through the bounds on $\|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\|$ and $\|\bar{\mathbf{x}}_{t-1} - \mathbf{y}_{t-1}^n\|$ in the proof of Theorem 1. We can eventually bound $\|\mathbf{x}_{t-1}^n - \mathbf{y}_{t-1}^n\|$ through the bound on $\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|$ established already in Lemma 3.

E. Strongly Convex Case

We further consider the case of strongly convex loss functions as in [26], [27], [54].

Assumption 5. (Strong Convexity) The local loss functions $f_t^n(\mathbf{x})$ are strongly convex over $\mathcal{X}$, i.e., $\exists \mu > 0$, such that for any $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, t , and n

$$f_t^n(\mathbf{y}) \geq f_t^n(\mathbf{x}) + \langle \nabla f_t^n(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \mu \|\mathbf{y} - \mathbf{x}\|^2. \quad (30)$$

Using the results in Lemma 4 and Theorem 1, the following theorem provides a network static regret bound for COLDQ with Assumption 5. As will be shown later in Section VI, the resulting network static regret is improved to $\mathcal{O}(\log T)$.

Theorem 5. (Network Static Regret Bound for Strongly Convex Loss) Under Assumptions 1-5, the network static regret of the COLDQ algorithm is upper bounded by

$$\begin{aligned} \text{REG}_s(T) &\leq \frac{1}{N} \sum_{n=1}^N \sum_{t=3}^T (\alpha_{t-1} - \alpha_{t-2} - \mu) \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \\ &\quad + \frac{\varepsilon_1^2}{4} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} + 2\alpha_1 R^2 + \varepsilon_1 R, \end{aligned} \quad (31)$$

where ε_1 is given in Theorem 1.

Proof: See Appendix I in the supplementary materials. ■

VI. DISCUSSION ON PERFORMANCE BOUNDS

With the general network dynamic regret, network static regret, and hard constraint violation bounds obtained in Section V, we now discuss how to set the algorithm parameters $\alpha_t, \eta_t, \gamma_t$ to yield the best performance bounds of COLDQ in terms of the growth rate over T . We further discuss the performance metrics of COLDQ over the current best results.

A. Time-Varying Constraints ($V_g > 0$)

From Theorems 1, 3, and 4, we can derive the following corollary on the network dynamic regret (14) and hard constraint violation (16) of COLDQ for time-varying constraints, i.e., $V_g > 0$ in (18).

Corollary 1. (Network Dynamic Regret #1) Under Assumptions 1-4, for any $V_x > 0$, and $V_g > 0$, let $\alpha_t = t^{\frac{1-V_x}{2}}$, $\eta_t = t^{-\varrho}$ and $\gamma_t = \epsilon t^\varrho$, where $\varrho > \frac{3-V_x}{2}$ and $\epsilon \in (0, G)$, COLDQ achieves

$$\text{REG}_d(T) = \mathcal{O}(T^{\frac{1+V_x}{2}}), \quad (32)$$

$$\text{VIO}_h(T) = \mathcal{O}(\min\{T^{\frac{3+V_x}{4}}, T^{V_g}\}). \quad (33)$$

Proof: See Appendix J in the supplementary materials. ■

We can see from Corollary 1 that COLDQ *simultaneously* achieves sublinear network dynamic regret and sublinear hard constraint violation, when the variation of the dynamic online global solution benchmark $\{\mathbf{x}_t^*\}$ in (17) and the variation of the local constraint function $\{\mathbf{g}_t^n\}$ in (18) grow sublinearly, i.e., $V_x, V_g < 1$. Otherwise, if the system varies adversarially, no OCO algorithm can simultaneously achieve sublinear dynamic regret and sublinear constraint violation, due to the lack of the current information of both the loss function and the constraint function in decision making [33], [60], [62]. Furthermore, all the algorithm parameters α_t, η_t , and γ_t of COLDQ are *independent* of the total time horizon T , allowing COLDQ to run arbitrarily long over time. This is in contrast to the hard-constrained OCO algorithms in [26], [28], [53] that can only run over a pre-determined finite time horizon T .

As summarized in Table II, for OCO with time-varying constraints, the algorithm in [27] is the only one that can simultaneously provide a dynamic regret bound and a hard constraint violation bound. The $\mathcal{O}(T^{\frac{1}{2}+V_x})$ dynamic regret and $\mathcal{O}(T^{\frac{3}{4}})$ hard constraint violation in [27] can be improved to $\mathcal{O}(T^{\frac{1+V_x}{2}})$ and $\mathcal{O}(T^{\frac{3+V_x}{4}})$ with the knowledge of V_x . However, the analysis in [27] is limited to *centralized* constrained OCO. Compared with [27], COLDQ provides *matched* $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret and *improved* $\mathcal{O}(\min\{T^{\frac{3+V_x}{4}}, T^{V_g}\})$ hard constraint violation, even under the *decentralized* setting. For the first time, the two bounds *smoothly* approach to the best-known $\mathcal{O}(T^{\frac{1}{2}})$ regret and $\mathcal{O}(1)$ violation under *both* centralized and decentralized settings, as the dynamics of the losses and constraints diminish, i.e., $V_x, V_g \rightarrow 0$.

From Theorems 2, 3, and 4, we can derive the following corollary on the network static regret (12) and hard constraint violation (16) of COLDQ for time-varying constraints. The proof follows from setting $V_x = 0$ in the proof of Corollary 1 and is omitted.

Corollary 2. (Network Static Regret #1) Under Assumptions 1-4, for any $V_g > 0$, let $\alpha_t = t^{\frac{1}{2}}$, $\eta_t = t^{-\varrho}$ and $\gamma_t = \epsilon t^\varrho$, where $\varrho > \frac{3}{2}$ and $\epsilon \in (0, G)$, COLDQ achieves

$$\text{REG}_s(T) = \mathcal{O}(T^{\frac{1}{2}}), \quad (34)$$

$$\text{VIO}_h(T) = \mathcal{O}(\min\{T^{\frac{3}{4}}, T^{V_g}\}). \quad (35)$$

We can see from Corollary 2 that COLDQ provides *matched* $\mathcal{O}(T^{\frac{1}{2}})$ static regret, compared with [27], [28], [54]. In the

decentralized setting, the $\mathcal{O}(\min\{T^{\frac{3}{4}}, T^{V_g}\})$ hard constraint violation of COLDQ is *no worse* than the current best $\mathcal{O}(T^{\frac{3}{4}})$ violation in [54] for any $V_g > 0$, and can be *strictly better* when $V_g < \frac{3}{4}$. In the *centralized* setting, the hard constraint violation bound of COLDQ can be better than the current best $\mathcal{O}(T^{\frac{1}{2}}(\log T)^{\frac{1}{2}})$ violation in [28] if $V_g \leq \frac{1}{2}$, i.e., when the constraint functions vary slowly, and worse otherwise. We note here that the constraint violation bound in [28] only holds with the static regret under the centralized setting. In contrast, the constraint violation bound of COLDQ holds for both the dynamic and static regret under the decentralized setting. Furthermore, the online algorithm in [28] requires specific values of the bounds D_1 and D_2 on the gradients of the loss and constraint functions to run. In addition, its gradient descent step size requires calculating $\|\nabla f_t^n(\mathbf{x}_t^n)\|^2$ and $\|\nabla \mathbf{g}_t^n(\mathbf{x}_t^n)\|^2$ over $t = 1, \dots, T$, which can be of high computational complexity especially for large-scale learning problems.

B. Fixed Constraints ($V_g = 0$)

From Theorems 1 and 3, we can derive the following corollary on the network dynamic regret and hard constraint violation of COLDQ for fixed constraints, i.e., $V_g = 0$.

Corollary 3. (Network Dynamic Regret #2) Under Assumptions 1-4, for any $V_x > 0$, and $V_g = 0$, let $\alpha_t = t^{\frac{1-V_x}{2}}$, $\eta_t = t^{-\varrho}$ and $\gamma_t = \epsilon t^{\varrho}$, where $\varrho > \frac{3-V_x}{2}$ and $\epsilon \in (0, G)$, COLDQ achieves

$$\text{REG}_d(T) = \mathcal{O}(T^{\frac{1+V_x}{2}}), \quad (36)$$

$$\text{VIO}_h(T) = \mathcal{O}(1). \quad (37)$$

Proof: We conducted a unified constraint violation analysis that is applicable to both the time-varying and fixed constraints in Theorem 3. Setting $V_g = 0$ in the proof of Corollary 1, we can show that Corollary 3 readily holds. ■

To the best of our knowledge, no existing decentralized constrained OCO algorithm has simultaneously provided a network dynamic regret bound and a hard constraint violation bound for fixed constraints. As summarized in Table I, the $\mathcal{O}(T^{\frac{1+V_x}{4}})$ and $\mathcal{O}(\log T)$ hard constraint violation bounds in [26] and [27] only hold for *centralized* constrained OCO. In contrast, Corollary 3 shows that COLDQ can achieve *matched* $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret and *optimal* $\mathcal{O}(1)$ hard constraint violation even under the *decentralized* setting.

From Theorems 2 and 3, we can derive the following corollary on the network static regret and hard constraint violation of COLDQ for fixed constraints. The proof follows from setting $V_g = 0$ in the proof of Corollary 2 and is omitted.

Corollary 4. (Network Static Regret #2) Under Assumptions 1-4, for $V_g = 0$, let $\alpha_t = t^{\frac{1}{2}}$, $\eta_t = t^{-\varrho}$ and $\gamma_t = \epsilon t^{\varrho}$, where $\varrho > \frac{3}{2}$ and $\epsilon \in (0, G)$, COLDQ achieves

$$\text{REG}_s(T) = \mathcal{O}(T^{\frac{1}{2}}), \quad (38)$$

$$\text{VIO}_h(T) = \mathcal{O}(1). \quad (39)$$

Under $\mathcal{O}(T^{\frac{1}{2}})$ static regret, the current best hard constraint violation bounds are $\mathcal{O}(T^{\frac{3}{4}})$ in [53] and $\mathcal{O}(1)$ in [27], for

decentralized and centralized constrained OCO, respectively. To the best of our knowledge, COLDQ is the *first* to simultaneously achieve the *best-known* $\mathcal{O}(T^{\frac{1}{2}})$ regret and $\mathcal{O}(1)$ constraint violation for *decentralized* constrained OCO.

Finally, in Corollaries 1 and 3, we require prior knowledge of the variation of the dynamic online global solution benchmark V_x to achieve $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret. To the best of our knowledge, whether $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret can be achieved for *decentralized* OCO without knowing V_x remains an open problem [63].⁴

C. Strongly Convex Loss

For standard OCO with only short-term constraints in the feasible set, an improved $\mathcal{O}(T^{V_x})$ dynamic regret can be achieved for strongly convex and smooth loss functions [15]. For soft-constrained OCO, the *centralized* online algorithm in [36] achieved $\mathcal{O}(T^{V_x})$ dynamic regret by performing multi-step gradient descent and then solving a per-slot problem. To the best of our knowledge, no existing constrained OCO work has shown such improvement by performing single-step gradient descent, or solving a single per-slot problem as in COLDQ. We therefore focus on the static regret for strongly convex loss functions in the following corollaries.⁵

From Theorems 3, 4, and 5, we can derive the following corollary on the static regret and hard constraint violation of COLDQ for strongly convex losses and time-varying constraints, i.e., $V_g > 0$.

Corollary 5. (Time-Varying Constraints) Under Assumptions 1-5, for any $V_g > 0$, let $\alpha_t = \mu t$, $\eta_t = t^{-\varrho}$ and $\gamma_t = \epsilon t^{\varrho}$, where $\varrho > 2$ and $\epsilon \in (0, G)$, COLDQ achieves

$$\text{REG}_s(T) = \mathcal{O}(\log T), \quad (40)$$

$$\text{VIO}_h(T) = \mathcal{O}(\min\{T^{\frac{1}{2}}(\log T)^{\frac{1}{2}}, T^{V_g}\}). \quad (41)$$

Proof: See Appendix K in the supplementary materials. ■

As summarized in Table II, under the *centralized* setting, the current best static regret and hard constraint violation bounds for strongly convex loss functions are $\mathcal{O}(\log T)$ and $\mathcal{O}(T^{\frac{1}{2}}(\log T)^{\frac{1}{2}})$ [27], [28]. Corollary 5 shows that COLDQ can provide *matched* $\mathcal{O}(\log T)$ static regret and *improved* $\mathcal{O}(\min\{T^{\frac{1}{2}}(\log T)^{\frac{1}{2}}, T^{V_g}\})$ hard constraint violation for strongly convex loss functions, even under the *decentralized* setting. The decentralized constrained OCO algorithm in [54] achieved $\mathcal{O}(T^c)$ static regret and $\mathcal{O}(T^{1-\frac{c}{2}})$ hard constraint violation for strongly convex loss functions. However, matching the $\mathcal{O}(T^c)$ static regret to $\mathcal{O}(\log T)$ necessities $c \rightarrow 0$, making the $\mathcal{O}(T^{1-\frac{c}{2}})$ hard constraint violation almost linear.

⁴COLDQ can achieve $\mathcal{O}(T^{\frac{1+V_x}{2}})$ dynamic regret in the *centralized* setting *without* prior knowledge of V_x , by maintaining multiple experts and aggregating their decisions based on the *loss-dependent* weights to track the optimal step size [11], [12], [14], [26], [27]. In the decentralized setting, the lack of accurate global loss function at the local learners makes the aggregation weights inaccurate for tracking the optimal step size.

⁵The same dynamic regret bounds in Corollaries 1 and 3 also hold for strongly convex functions. Leveraging additional assumptions or gradient descent steps to improve the dynamic regret bound is not the focus of this paper. We refer interested readers to [15]-[17].

From Theorems 3 and 5, we can derive the following corollary on the network static regret and hard constraint violation of COLDQ for strongly convex losses and fixed constraints, *i.e.*, $V_g = 0$. The proof follows from letting $V_g = 0$ in the proof of Corollary 5 and is omitted.

Corollary 6. (Fixed Constraints) Under Assumptions 1-5, for $V_g = 0$, let $\alpha_t = \mu t$, $\eta_t = t^{-\varrho}$ and $\gamma_t = \epsilon t^\varrho$, where $\varrho > 2$ and $\epsilon \in (0, G)$, COLDQ achieves

$$\text{REG}_s(T) = \mathcal{O}(\log T), \quad (42)$$

$$\text{VIO}_h(T) = \mathcal{O}(1). \quad (43)$$

Under $\mathcal{O}(\log T)$ static regret, the current best hard constraint violation bounds are $\mathcal{O}(T^{\frac{1}{2}}(\log T)^{\frac{1}{2}})$ in [26] and $\mathcal{O}(1)$ in [27], for decentralized and centralized constrained OCO with strongly convex losses, respectively. To the best of our knowledge, COLDQ is the *first* to simultaneously achieve the *best-known* $\mathcal{O}(\log T)$ regret and $\mathcal{O}(1)$ constraint violation for *decentralized* constrained OCO with strongly convex losses.

VII. SIMULATION RESULTS

In this section, we first apply COLDQ to decentralized constrained online quadratic programming, with both time-varying and fixed constraints using synthetic data. We then apply COLDQ to communication-efficient decentralized learning, for both convex logistic regression and non-convex neural network training using canonical image classification datasets.

We consider a decentralized learning system with $N = 20$ learners communicating over a time-varying network with undirected edges [48], [50], [54]. At each time t , each edge (k, n) between learners k and n is established randomly with 0.1 probability. We set the edge weight as $p_t^{n,k} = \frac{1}{N}$ if (k, n) exists, and 0 otherwise, so that Assumption 4.1 holds. We set $p_t^{n,n} = 1 - \sum_{k \in \mathcal{N}, k \neq n} p_t^{n,k}$ to ensure the doubly stochastic Assumption 4.2 holds. Additionally, we consider the edge between learner n and $n + 1$ always exists, such that the strongly connected Assumption 4.3 holds.⁶

A. Decentralized Constrained Online Quadratic Programming

For online quadratic programming, we compare COLDQ with the following hard-constrained OCO works:

- [26]: A *centralized* OCO algorithm that achieves $\mathcal{O}(\frac{1+V_g}{4})$ violation for *fixed* constraints.
- [27]: A *centralized* virtual-queue-based OCO algorithm that achieves $\mathcal{O}(T^{\frac{3+V_g}{4}})$ and $\mathcal{O}(1)$ violations for time-varying and fixed constraints, respectively.
- [28]: A *centralized* projection-based OCO algorithm that achieves $\mathcal{O}(T^{\frac{1}{2}} \log T)$ violation for time-varying constraints.
- [53]: A decentralized projection-based OCO algorithm that achieves $\mathcal{O}(T^{\frac{3}{4}})$ violation for *fixed* constraints.
- [54]: A decentralized OCO algorithm that achieves $\mathcal{O}(T^{\frac{3+V_g}{4}})$ violation for time-varying constraints.

Among the above benchmarks, only [27] is designed for both time-varying and fixed constraints. The algorithms in

[26]-[28] are designed for centralized systems. We implemented these algorithms in the decentralized setting with the same aggregation method as COLDQ. In the following two experiments, we use the accumulated loss $f(T) = \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T f_t(\mathbf{x}_t^n)$ and hard constraint violation $g(T) = \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^T \|\mathbf{g}_t^n(\mathbf{x}_t^n)\|$ as our performance metrics. For fair comparison among the above benchmarks, we have fine-tuned their suggested parameters to achieve the closest possible accumulated loss. We have provided public access to our code at <https://github.com/yituo-liu/TON26-COLDQ>.

1) *Experiment on Time-Varying Constraints:* We extend the centralized time-varying constrained online quadratic programming problem in [27], [28] to the decentralized setting. We set the local loss function as $f_t^n(\mathbf{x}) = \frac{1}{2} \|\mathbf{H}_t^n \mathbf{x} - \mathbf{o}_t^n\|^2$, where $\mathbf{H}_t^n \in \mathbb{R}^{4 \times 10}$, $\mathbf{x} \in \mathbb{R}^{10}$, and $\mathbf{o}_t^n \in \mathbb{R}^4$. Each element of $\mathbf{H}_t^n$ is uniformly distributed, *i.e.*, $H_t^{n,i,j} \sim U(-1, 1), \forall i, j$. Each element of $\mathbf{o}_t^n$ satisfies $o_t^{n,i} = \sum_{j=1}^{10} H_t^{n,i,j} + \epsilon^{n,i}$, where $\epsilon^{n,i} \sim N(0, 1), \forall i$. We set the local constraint functions as $\mathbf{g}_t^n(\mathbf{x}) = \mathbf{A}_t^n \mathbf{x} - \mathbf{b}_t^n$, where $\mathbf{A}_t^n \in \mathbb{R}^{2 \times 10}$ and $\mathbf{b}_t^n \in \mathbb{R}^2$. We generate $A_t^{n,i,j} \sim U(0, 1), \forall i, j$ and $b_t^{n,i} \sim U(0, 1), \forall i$. We set the feasible set as $\mathcal{X} = \{\mathbf{x} \mid -5 \preceq \mathbf{x} \preceq 5\}$.

We compare COLDQ with the state-of-the-art time-varying constrained OCO algorithms in [27], [28] and [54]. Fig. 1 shows that COLDQ can achieve over 40% lower hard constraint violation than [27], without sacrificing any accumulated loss. Furthermore, the performance gain of COLDQ over [28], [54] is more substantial. This is because the doubly-bounded queue and the direct minimization of the constraint function allow COLDQ to more strictly control the constraint violation.

2) *Experiment on Fixed Constraints:* We also extend the centralized online quadratic programming problem with fixed constraints in [26], [27] to the decentralized setting. We set the local loss function as $f_t^n(\mathbf{x}) = \|\mathbf{x} - \boldsymbol{\theta}_t^n\|^2 + 20(\boldsymbol{\theta}_t^n, \mathbf{x})$, where $\boldsymbol{\theta}_t^n = \boldsymbol{\theta}_t^{n,1} + \boldsymbol{\theta}_t^{n,2} + \boldsymbol{\theta}_t^{n,3} \in \mathbb{R}^2$ and $\mathbf{x} \in \mathbb{R}^2$. Each element in $\boldsymbol{\theta}_t^{n,1}, \boldsymbol{\theta}_t^{n,2}, \boldsymbol{\theta}_t^{n,3}$ are separately set as $\theta_t^{n,1,j} \sim U(-t^{1/10}, t^{1/10}), \forall j$; $\theta_t^{n,2,j} \sim U(-1, 0), \forall j$ for $t \in [1, 300] \cup [400, 700] \cup [800, 1000]$, and $\theta_t^{n,2,j} \sim U(0, 1), \forall j$ otherwise; and $\theta_t^{n,3,j} = (-1)^{\mu_t}, \forall j$ with the sequence of μ_t being a random permutation of the vector $[1 : 1000]$. We set the local constraint function as $\mathbf{g}^n(\mathbf{x}) = \mathbf{A}^n \mathbf{x} - \mathbf{b}^n$, where $\mathbf{A}^n \in \mathbb{R}^{3 \times 2}$ and $\mathbf{b}^n \in \mathbb{R}^3$ with $A^{n,i,j} \sim U(0, 0.5), \forall i, j$, and $b^{n,i} \sim U(0, 0.3), \forall i$ and the feasible set as $\mathcal{X} = \{\mathbf{x} \mid -1 \preceq \mathbf{x} \preceq 1\}$.

We compare COLDQ with the state-of-the-art fixed constrained OCO algorithms in [26], [27], and [53]. We can see from Fig. 2 that COLDQ demonstrates significant reductions in both the accumulated loss and the hard constraint violation. In the following experiments, we compare COLDQ with [27], since it yields better performance than [26], [28], [53], [54].

B. Communication Efficient Decentralized Learning

To further evaluate the performance of COLDQ in practical systems, we apply it to decentralized learning with quantization. For efficient communication, after obtaining the local decision $\mathbf{x}_t^n$, each learner n generates a quantized local decision $\mathcal{C}(\mathbf{x}_t^n)$, where $\mathcal{C}(\mathbf{x})$ is a general quantization operator. In our experiment, the i -th element of a quantized decision $\mathcal{C}(x^i)$ is generated as $x_{\max} \cdot \text{sgn}(x^i) \cdot \phi(x^i)$, where $x_{\max}$ is the

⁶Optimizing the edge weights for decentralized learning is out of the scope of this work. We refer interested readers to the recent work [64].

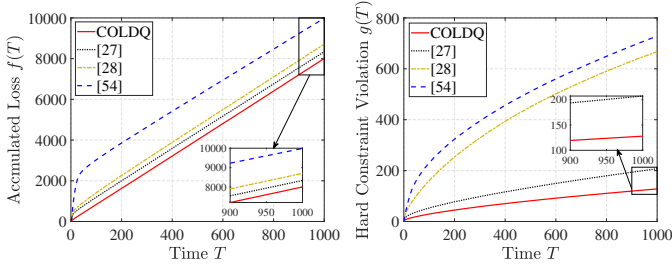


Fig. 1: Experiment on time-varying constraints.

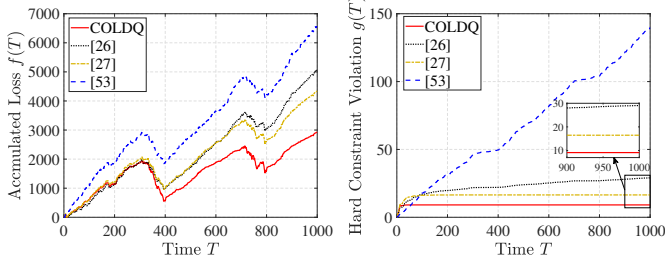


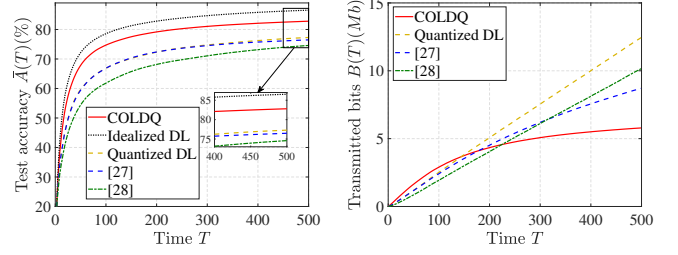
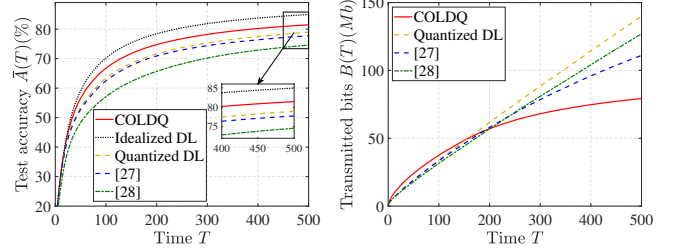
Fig. 2: Experiment on fixed constraints.

maximum decision value, $\text{sgn}(x)$ returns the sign of x , and $\phi(x) = \frac{l}{s}$ with probability $1 - \frac{|x|}{x_{\max}}s + l$ and $\phi(x) = \frac{l+1}{s}$ otherwise, with $0 \leq l < s - 1$ being an integer that satisfies $\frac{|x|}{x_{\max}} \in [\frac{l}{s}, \frac{l+1}{s}]$ and $s = 2^b$ being the quantization level [65]. Each learner n then aggregates the quantized local decisions received from its neighbors as $\mathbf{y}_t^n = \sum_{k=1}^N p_t^{n,k} \mathcal{C}(\mathbf{x}_t^k)$.

Most existing quantization operators can guarantee unbiased quantization, i.e., $\mathbb{E}\{\mathcal{C}(\mathbf{x})\} = \mathbf{x}$. However, as shown in [65]–[67], the actual learning performance largely depends on the upper bound on the quantization variance $\mathbb{E}\{\|\mathbf{x} - \mathcal{C}(\mathbf{x})\|^2\}$. To optimize the later quantization performance during the local decision updating, we design the constraint function as $g_t^n(\mathbf{x}) = \|\mathbf{x} - \mathcal{C}(\mathbf{x}_t^n)\|^2 - \sigma$, where $\sigma > 0$ is a limit on the quantization variance. Note that the information of $\mathcal{C}(\mathbf{x}_t^n)$ is only available after learner n updates its local decision $\mathbf{x}_t^n$, i.e., only the delayed constraint function $g_{t-1}^n(\mathbf{x}) = \|\mathbf{x} - \mathcal{C}(\mathbf{x}_{t-1}^n)\|^2 - \sigma$ is available for learner n to update its local decision $\mathbf{x}_t^n$. As the learning process converges, the difference between two adjacent quantized local decisions $\mathcal{C}(\mathbf{x}_{t-1}^n) - \mathcal{C}(\mathbf{x}_t^n)$ diminishes, $g_{t-1}^n(\mathbf{x})$ becomes a more accurate estimation of $g_t^n(\mathbf{x})$. To further reduce the communication overhead, we use conditional entropy coding [68] to only transmit the information in the difference between $\mathcal{C}(\mathbf{x}_t^n)$ and $\mathcal{C}(\mathbf{x}_{t-1}^n)$. The incurred transmitted bits $H(\mathcal{C}(\mathbf{x}_t^n)|\mathcal{C}(\mathbf{x}_{t-1}^n))$ can be much lower than $H(\mathcal{C}(\mathbf{x}_t^n))$ using entropy coding, where $H(\mathcal{C}(\mathbf{x}_t^n))$ is the marginal entropy of $\mathcal{C}(\mathbf{x}_t^n)$ and $H(\mathcal{C}(\mathbf{x}_t^n)|\mathcal{C}(\mathbf{x}_{t-1}^n))$ is the conditional entropy of $\mathcal{C}(\mathbf{x}_t^n)$ given $\mathcal{C}(\mathbf{x}_{t-1}^n)$.

For communication-efficient decentralized learning, we compare COLDQ with the following schemes:

- *Idealized DL*: We alternate local update $\mathbf{x}_t^n = \mathbf{y}_{t-1}^n - \frac{1}{2\alpha} \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n)$ and local aggregation $\mathbf{y}_t^n = \sum_{k=1}^N p_t^{n,k} \mathbf{x}_t^k$ in (3) for each learner n . It represents the *idealized* decentralized learning (DL) algorithm without any limit on the communication resources.
- *Quantized DL*: We adopt the standard quantized DL method in [69], [70]. It updates the local models via (23)

Fig. 3: Test accuracy $\bar{A}(T)$ and transmitted bits $B(T)$ vs. time T for convex logistic regression.Fig. 4: Test accuracy $\bar{A}(T)$ and transmitted bits $B(T)$ vs. time T for non-convex neural network training.

and then use the *same* quantization method as COLDQ to generate $\mathcal{C}(\mathbf{x}_t^n)$.

- [27]: We extend the *centralized* virtual-queue-based OCO algorithm to DL with the *same* quantization and aggregation methods as COLDQ.
- [28]: We extend the *centralized* projection-based OCO algorithm to DL with the *same* quantization and aggregation methods as COLDQ.

We conduct experiments over non-i.i.d. datasets, where 20% of the local training samples belong to one label, and the remaining 80% belong to the other labels [71]–[73].

1) *Convex Logistic Regression*: We first use the MNIST dataset [74]. Its training dataset $\mathcal{D}$ and test dataset $\mathcal{S}$ consist of 6×10^4 and 1×10^4 data samples, respectively. Each data sample $(\mathbf{u}, v)$ represents an image with 28×28 pixels and $J = 10$ possible labels. At each time t , each learner n randomly selects $|\mathcal{D}_t^n| = 32$ training samples from $\mathcal{D}$. We use the cross-entropy loss for multinomial logistic regression, given by $l(\mathbf{x}; \mathbf{u}, v) = -\sum_{j=1}^J \mathbf{1}\{v = j\} \log \frac{\exp(\langle \mathbf{x}[j], \mathbf{u} \rangle)}{\sum_{k=1}^J \exp(\langle \mathbf{x}[k], \mathbf{u} \rangle)}$, where $\mathbf{x} = [\mathbf{x}[1]^\top, \dots, \mathbf{x}[J]^\top]^\top$ with $\mathbf{x}[j] \in \mathbb{R}^{784}$ being the model for label j . The entire model $\mathbf{x}$ is of dimension $d = 7840$. We then define the local loss function as $f_t^n(\mathbf{x}) = \frac{1}{|\mathcal{D}_t^n|} \sum_{i=1}^{|\mathcal{D}_t^n|} l(\mathbf{x}; \mathbf{u}_t^{n,i}, v_t^{n,i})$. Our performance metrics are the time-averaged test accuracy $\bar{A}(T) = \frac{1}{NT|\mathcal{S}|} \sum_{t=1}^T \sum_{n=1}^N \sum_{i=1}^{|\mathcal{S}|} \mathbf{1}\{\arg \max_j \frac{\exp(\langle \mathbf{x}_t^n[j], \mathbf{u}^i \rangle)}{\sum_{k=1}^J \exp(\langle \mathbf{x}_t^n[k], \mathbf{u}^i \rangle)} = v^i\}$ and the total number of transmitted bits $B(T) = \sum_{t=1}^T \sum_{n=1}^N H(\mathcal{C}(\mathbf{x}_t^n)|\mathcal{C}(\mathbf{x}_{t-1}^n))$.

Fig. 3 shows $\bar{A}(T)$ and $B(T)$ versus time T . We set the same α_t , σ , and $x_{\max}$ for all schemes. We have fine-tuned the other hyper-parameters to reach the closest possible total transmitted bits. We observe that the test accuracy achieved by COLDQ is close to Idealized DL. Compared with Quantized DL, [27] and [28], COLDQ uses over 30% less transmitted bits, while achieving at least 5% higher test accuracy.

2) *Non-Convex Neural Network Training*: While the performance bounds of COLDQ derived in Sections V and VI are for convex loss functions, COLDQ can be implemented with non-convex losses. We study the performance of COLDQ on CIFAR-10 [75] image classification using a pre-trained ResNet-20 [76]. The total number of model parameters is 272,474. To speed up the training process, we only train the second convolutional layer in the last residual stage, which has 36,864 model parameters. In Fig. 4, we compare the averaged test accuracy and the total transmitted bits among CODLQ, Quantized DL, [27], and [28]. We observe that the total transmission bits yielded by all schemes are higher than the previous convex logistic regression scenario, due to the larger number of model parameters. Similar to Fig. 3, CODLQ achieves a test accuracy that is close to Idealized DL, while requiring much less transmitted bits than Quantized DL, [27], and [28].

VIII. CONCLUSIONS

We propose an effective COLDQ algorithm for decentralized OCO with time-varying constraints. We design a novel virtual queue that is bounded both from above and below to strictly control the hard constraint violation. Through a new penalized Lyapunov drift analysis, COLDQ provides $\mathcal{O}(T^{\frac{1+V_x}{2}})$ network dynamic regret and $\mathcal{O}(\min\{T^{\frac{3+V_x}{4}}, T^{V_g}\})$ hard constraint violation. For the first time, the two bounds smoothly approach to the best-known $\mathcal{O}(T^{\frac{1}{2}})$ regret and $\mathcal{O}(1)$ violation, as the dynamics of the losses and constraints represented by V_x and V_g diminish. We further study the case of strongly-convex loss functions, and demonstrate that COLDQ provides $\mathcal{O}(\log T)$ static regret $\mathcal{O}(\min\{T^{\frac{1}{2}}(\log T)^{\frac{1}{2}}, T^{V_g}\})$ hard constraint violation. Experimental results on synthetic and canonical datasets for both convex and non-convex losses complement our theoretical analysis.

ACKNOWLEDGEMENT

In this paper, Y. Liu, W. Qin, and J. Wang were supported by the Hong Kong Research Grants Council (RGC) Early Career Scheme (ECS) under Grant 22200324 and Research Matching Grant Scheme (RMGS), the National Natural Science Foundation of China (NSFC) Young Scientists Fund under Grant 62502407, the Department of Science and Technology of Guangdong Province (GDSTC) General Program under Grant 2026A151010105, Guangdong and Hong Kong Universities “1+1+1” Joint Research Collaboration Scheme, and Huawei Technologies Co., Ltd., China. A preliminary version of this paper was published as [1] J. Wang, B. Yan, and Y. Liu, “Doubly bounded queue for constrained online learning: Keeping pace with dynamics of both loss and constraint,” in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2025. We acknowledge that B. Yan has contributed to the preliminary conference paper [1]. Also, we acknowledge that the preliminary conference paper was supported in part by the Hong Kong RGS ECS under grant 22200324.

REFERENCES

- [1] J. Wang, B. Yan, and Y. Liu, “Doubly-bounded queue for constrained online learning: Keeping pace with dynamics of both loss and constraint,” in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2025.
- [2] S. Shalev-Shwartz, “Online learning and online convex optimization,” *Found. Trends Mach. Learn.*, vol. 4, no. 2, pp. 107–194, 2012.
- [3] E. Hazan, “Introduction to online convex optimization,” *Foundations and Trends in Optimization*, vol. 2, no. 3–4, pp. 157–325, 2016.
- [4] S. Balseiro, H. Lu, and V. Mirrokni, “Dual mirror descent for online allocation problems,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020.
- [5] W.-K. Hsu, J. Xu, X. L. Lin, and M. R. Bell, “Integrated online learning and adaptive control in queueing systems with uncertain payoffs,” *Operations Research*, vol. 70, no. 2, pp. 1166–1181, 2021.
- [6] M. Shi, X. Lin, and S. Fahmy, “Competitive online convex optimization with switching costs and ramp constraints,” *IEEE/ACM Trans. Netw.*, vol. 29, no. 2, pp. 876–889, 2021.
- [7] J. Wang, B. Liang, M. Dong, G. Boudreau, and A. Afana, “Age-of-information minimization with weight limits for semi-asynchronous online distributed optimization,” *IEEE Trans. Netw.*, vol. 33, no. 6, pp. 3162–3178, 2025.
- [8] M. Zinkevich, “Online convex programming and generalized infinitesimal gradient ascent,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2003.
- [9] E. Hazan, A. Agarwal, and S. Kale, “Logarithmic regret algorithms for online convex optimization,” *Mach. Learn.*, vol. 69, no. 2, pp. 169–192, 2007.
- [10] E. C. Hall and R. M. Willett, “Online convex optimization in dynamic environments,” *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 4, pp. 647–662, 2015.
- [11] A. Jadbabaie, A. Rakhlin, S. Shahrampour, and K. Sridharan, “Online optimization: Competing with dynamic comparators,” in *Proc. Int. Conf. Artif. Intell. Stat. (AISTATS)*, 2015.
- [12] L. Zhang, S. Lu, and Z.-H. Zhou, “Adaptive online learning in dynamic environments,” in *Proc. Adv. Neural Info. Proc. Sys. (NeurIPS)*, 2018.
- [13] N. Eshraghi and B. Liang, “Distributed online optimization over a heterogeneous network with any-batch mirror descent,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2020.
- [14] P. Zhao, Y.-J. Zhang, L. Zhang, and Z.-H. Zhou, “Adaptivity and non-stationarity: Problem-dependent dynamic regret for online convex optimization,” *J. Mach. Learn. Res.*, vol. 25, no. 98, pp. 1–52, 2024.
- [15] A. Mokhtari, S. Shahrampour, A. Jadbabaie, and A. Ribeiro, “Online optimization in dynamic environments: Improved regret rates for strongly convex problems,” in *Proc. IEEE Conf. Decis. Control. (CDC)*, 2016.
- [16] P. Zhao and L. Zhang, “Improved analysis for dynamic regret of strongly convex and smooth functions,” in *Proc. Learn. Dyn. Control (LADC)*, 2021.
- [17] D. Baby and Y.-X. Wang, “Optimal dynamic regret in exp-concave online learning,” in *Proc. Conf. Learn. Theory (COLT)*, 2021.
- [18] Z. Liu, I. Liu, S. Low, and A. Wierman, “Pricing data center demand response,” *ACM SIGMETRICS Perform. Eval. Rev.*, vol. 42, pp. 111–123, 2014.
- [19] Y. Zhang, M. H. Hajiesmaili, S. Cai, M. Chen, and Q. Zhu, “Peak-aware online economic dispatching for microgrids,” *IEEE Trans. Smart Grid*, vol. 9, pp. 323–335, 2018.
- [20] H. Yu and M. J. Neely, “Learning-aided optimization for energy-harvesting devices with outdated state information,” *IEEE/ACM Trans. Netw.*, vol. 27, pp. 1501–1514, 2019.
- [21] M. Mahdavi, R. Jin, and T. Yang, “Trading regret for efficiency: online convex optimization with long term constraints,” *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 2503–2528, 2012.
- [22] R. Jenatton, J. Huang, and C. Archambeau, “Adaptive algorithms for online convex optimization with long-term constraints,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2016.
- [23] H. Yu and M. J. Neely, “A low complexity algorithm with $\mathcal{O}(\sqrt{T})$ regret and $\mathcal{O}(1)$ constraint violations for online convex optimization with long term constraints,” *J. Mach. Learn. Res.*, vol. 21, no. 1, pp. 1–24, 2020.
- [24] J. Wang, M. Dong, B. Liang, and G. Boudreau, “Periodic updates for constrained OCO with application to large-scale multi-antenna systems,” *IEEE Trans. Mobile Comput.*, vol. 22, no. 11, pp. 6705–6722, 2023.
- [25] J. Yuan and A. Lamperski, “Online convex optimization for cumulative constraints,” in *Proc. Adv. Neural Info. Proc. Sys. (NeurIPS)*, 2018.
- [26] X. Yi, X. Li, T. Yang, L. Xie, T. Chai, and K. Johansson, “Regret and cumulative constraint violation analysis for online convex optimization with long term constraints,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021.

- [27] H. Guo, X. Liu, H. Wei, and L. Ying, "Online convex optimization with hard constraints: Towards the best of two worlds and beyond," in *Proc. Adv. Neural Info. Proc. Sys. (NeurIPS)*, 2022.
- [28] A. Sinha and R. Vaze, "Optimal algorithms for online convex optimization with adversarial constraints," in *Proc. Adv. Neural Info. Proc. Sys. (NeurIPS)*, 2024.
- [29] H. Yu, M. Neely, and X. Wei, "Online convex optimization with stochastic constraints," in *Proc. Adv. Neural Info. Proc. Sys. (NeurIPS)*, 2017.
- [30] X. Wei, H. Yu, and M. J. Neely, "Online primal-dual mirror descent under stochastic constraints," in *Proc. ACM Meas. Anal. Comput. Syst. (POMACS)*, 2020.
- [31] N. Liakopoulos, A. Destounis, G. Paschos, T. Spyropoulos, and P. Mertikopoulos, "Cautious regret minimization: Online optimization with long-term budget constraints," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019.
- [32] X. Cao, J. Zhang, and H. V. Poor, "Constrained online convex optimization with feedback delays," *IEEE Trans. Autom. Control*, vol. 66, no. 11, pp. 5049–5064, 2021.
- [33] T. Chen, Q. Ling, and G. B. Giannakis, "An online convex optimization approach to proactive network resource allocation," *IEEE Trans. Signal Process.*, vol. 65, no. 24, pp. 6350–6364, 2017.
- [34] X. Cao and K. R. Liu, "Online convex optimization with time-varying constraints and bandit feedback," *IEEE Trans. Autom. Control*, vol. 64, no. 7, pp. 2665–2680, 2019.
- [35] Q. Liu, W. Wu, L. Huang, and Z. Fang, "Simultaneously achieving sublinear regret and constraint violations for online convex optimization with time-varying constraints," *ACM SIGMETRICS Perform. Eval. Rev.*, vol. 49, no. 3, pp. 4–5, 2022.
- [36] J. Wang, M. Dong, B. Liang, G. Boudreau, and H. Abou-Zeid, "Delay-tolerant OCO with long-term constraints: Algorithm and its application to network resource allocation," *IEEE/ACM Trans. Netw.*, vol. 31, no. 1, pp. 147–163, 2023.
- [37] J. Lekeufack and M. I. Jordan, "An optimistic algorithm for online convex optimization with adversarial constraints," *arXiv:2412.08060*, 2024.
- [38] F. Yan, S. Sundaram, S. Vishwanathan, and Y. Qi, "Distributed autonomous online learning: Regrets and intrinsic privacy-preserving properties," *IEEE Trans. Knowl. Data Eng.*, vol. 25, no. 11, pp. 2483–2493, 2013.
- [39] A. Koppel, F. Y. Jakubiec, and A. Ribeiro, "A saddle point algorithm for networked online convex optimization," *IEEE Trans. Signal Process.*, vol. 63, no. 19, pp. 5149–5164, 2015.
- [40] S. Hosseini, A. Chapman, and M. Mesbahi, "Online distributed convex optimization on dynamic networks," *IEEE Trans. Autom. Control*, vol. 61, no. 11, pp. 3545–3550, 2016.
- [41] S. Shahrampour and A. Jadbabaie, "Distributed online optimization in dynamic environments using mirror descent," *IEEE Trans. Autom. Control*, vol. 63, no. 3, pp. 714–725, 2018.
- [42] N. Eshraghi and B. Liang, "Improving dynamic regret in distributed online mirror descent using primal and dual information," in *Learn. Dyn. Control (LADC)*, 2022.
- [43] W. Zhang, P. Zhao, W. Zhu, S. C. Hoi, and T. Zhang, "Projection-free distributed online learning in networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017.
- [44] Y. Wan, G. Wang, W.-W. Tu, and L. Zhang, "Projection-free distributed online learning with sublinear communication complexity," *J. Mach. Learn. Res.*, vol. 23, no. 172, pp. 1–53, 2022.
- [45] D. Yuan, D. W. Ho, and G.-P. Jiang, "An adaptive primal-dual subgradient algorithm for online distributed constrained optimization," *IEEE Trans. Cybern.*, vol. 48, no. 11, pp. 3045–3055, 2018.
- [46] X. Li, X. Yi, and L. Xie, "Distributed online optimization for multi-agent networks with coupled inequality constraints," *IEEE Trans. Autom. Control*, vol. 66, no. 8, pp. 3575–3591, 2021.
- [47] X. Cao, "Distributed constrained online convex optimization with adaptive quantization," *Automatica*, vol. 169, p. 111828, 2024.
- [48] X. Yi, X. Li, L. Xie, and K. H. Johansson, "Distributed online convex optimization with time-varying coupled inequality constraints," *IEEE Trans. Signal Process.*, vol. 68, pp. 731–746, 2020.
- [49] P. Sharma, P. Khanduri, L. Shen, D. J. Bucci, and P. K. Varshney, "On distributed online convex optimization with sublinear dynamic regret and fit," in *Proc. Asilomar Conf. Signals, Syst., Comput. (ACSSC)*, 2021.
- [50] X. Yi, X. Li, T. Yang, L. Xie, T. Chai, and K. H. Johansson, "Distributed bandit online convex optimization with time-varying coupled inequality constraints," *IEEE Trans. Autom. Control*, vol. 66, no. 10, pp. 4620–4635, 2021.
- [51] J. Li, C. Gu, Z. Wu, and T. Huang, "Online learning algorithm for distributed convex optimization with time-varying coupled constraints and bandit feedback," *IEEE Trans. Cybern.*, vol. 52, no. 2, pp. 1009–1020, 2022.
- [52] D. Yuan, A. Proutiere, and G. Shi, "Distributed online linear regressions," *IEEE Trans. Inf. Theory*, vol. 67, no. 1, pp. 616–639, 2021.
- [53] D. Yuan, A. Proutiere, and G. Shi, "Distributed online optimization with long-term constraints," *IEEE Trans. Autom. Control*, vol. 67, no. 3, pp. 1089–1104, 2022.
- [54] X. Yi, X. Li, T. Yang, L. Xie, T. Chai, and K. H. Johansson, "Regret and cumulative constraint violation analysis for distributed online constrained convex optimization," *IEEE Trans. Autom. Control*, vol. 68, no. 5, pp. 2875–2890, 2023.
- [55] J. Wang, M. Dong, B. Liang, G. Boudreau, and A. Afana, "Exploring temporal similarity for joint computation and communication in online distributed optimization," *IEEE Trans. Netw.*, vol. 33, no. 3, pp. 1309–1325, 2025.
- [56] J. Wang, Y. Liu, B. Liang, and M. Dong, "Constrained over-the-air model updating for wireless online federated learning with delayed information," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, 2025.
- [57] A. Nedic and A. Ozdaglar, "Distributed subgradient methods for multi-agent optimization," *IEEE Trans. Autom. Control*, vol. 54, no. 1, pp. 48–61, 2009.
- [58] B. Gharesifard and J. Cortés, "When does a digraph admit a doubly stochastic adjacency matrix?" in *Proc. Amer. Control Conf. (ACC)*, 2010.
- [59] S. Mannor, J. N. Tsitsiklis, and J. Y. Yu, "Online learning with sample path constraints," *J. Mach. Learn. Res.*, vol. 10, no. 3, pp. 569–590, 2009.
- [60] O. Besbes, Y. Gur, and A. Zeevi, "Non-stationary stochastic optimization," *Oper. Res.*, vol. 63, no. 5, pp. 1227–1244, 2015.
- [61] M. J. Neely, *Stochastic Network Optimization with Application on Communication and Queueing Systems*. Springer Nature, 2022.
- [62] X. Cao, J. Zhang, and H. V. Poor, "A virtual-queue-based algorithm for constrained online convex optimization with applications to data center resource allocation," *IEEE J. Sel. Top. Signal Process.*, vol. 12, no. 4, pp. 703–716, 2018.
- [63] H. Chen and Q. Sun, "Decentralized online riemannian optimization with dynamic environments," *arXiv:2410.05128*, 2024.
- [64] J. Wang, B. Liang, Z. Zhu, E. T. Fapi, and H. Dalal, "Communication-efficient network topology in decentralized learning: A joint design of consensus matrix and resource allocation," *IEEE Trans. Netw.*, vol. 33, no. 2, pp. 761–776, 2025.
- [65] D. Alistarh, D. Grubic, J. Li, R. Tomioka, and M. Vojnovic, "Qsgd: Communication-efficient sgd via gradient quantization and encoding," in *Proc. Adv. Neural Info. Proc. Sys. (NeurIPS)*, 2017.
- [66] J. Wangni, J. Wang, J. Liu, and T. Zhang, "Gradient sparsification for communication-efficient distributed optimization," in *Proc. Adv. Neural Info. Proc. Sys. (NeurIPS)*, 2018.
- [67] D. Kovalev, A. Koloskova, M. Jaggi, P. Richtarik, and S. Stich, "A linearly convergent algorithm for decentralized optimization: Sending less bits for free!" in *Int. Conf. Artif. Intell. Stat. (AISTATS)*, 2021.
- [68] D. Slepian and J. Wolf, "Noiseless coding of correlated information sources," *IEEE Trans. Inf. Theory*, vol. 19, no. 4, pp. 471–480, 1973.
- [69] M. G. Rabbat and R. D. Nowak, "Quantized incremental algorithms for distributed optimization," *IEEE J. Sel. Areas Commun.*, vol. 23, no. 4, pp. 798–808, 2005.
- [70] A. Nedic, A. Olshevsky, A. Ozdaglar, and J. N. Tsitsiklis, "Distributed subgradient methods and quantization effects," in *Proc. IEEE Conf. Decis. Control. (CDC)*, 2008.
- [71] H. Wang, Z. Kaplan, D. Niu, and B. Li, "Optimizing federated learning on non-iid data with reinforcement learning," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, 2020.
- [72] Y. Liao, Y. Xu, H. Xu, M. Chen, L. Wang, and C. Qiao, "Asynchronous decentralized federated learning for heterogeneous devices," *IEEE/ACM Trans. Netw.*, vol. 32, no. 5, pp. 4535–4550, 2024.
- [73] J. Liu, J. Liu, H. Xu, Y. Liao, Z. Wang, and Q. Ma, "Yoga: Adaptive layer-wise model aggregation for decentralized federated learning," *IEEE/ACM Trans. Netw.*, vol. 32, no. 2, pp. 1768–1780, 2024.
- [74] Y. LeCun, C. Cortes, and C. Burges, "The MNIST database," 1998.
- [75] A. Krizhevsky, G. Hinton *et al.*, "Learning multiple layers of features from tiny images," 2009.
- [76] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2016.



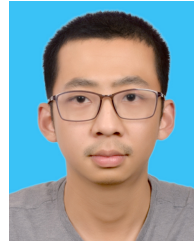
Yituo Liu (Student Member, IEEE) received the B.Eng. degree in electrical and information engineering from Xiamen University, Malaysia, in 2022, and the M.Sc. degree in communications engineering from Nanyang Technological University, Singapore, in 2024. He is currently pursuing the M.Phil. degree with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China. His research interests include online learning and distributed machine learning.



Juncheng Wang (Member, IEEE) received the B.Eng. degree in Electrical Engineering from Shanghai Jiao Tong University, Shanghai, China, in 2014, the M.Sc. degree in Electrical and Computer Engineering from the University of Alberta, Edmonton, AB, Canada, in 2017, and the Ph.D. degree in Electrical and Computer Engineering from the University of Toronto, Toronto, ON, Canada, in 2023. He is currently an Assistant Professor with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China. His research interests include network artificial intelligence, distributed machine learning, and online optimization.



Weiyi Qin (Student Member, IEEE) received the B.S. degree in information and computing science from Anhui University, Hefei, China, in 2022, and the M.Eng. degree in control science and engineering from the University of Science and Technology of China, Hefei, China, in 2025. He is currently pursuing the Ph.D. degree with the Department of Computer Science, Hong Kong Baptist University, Hong Kong, China. His research interests include online learning and distributed machine learning.



Jianxiong Guo (Member, IEEE) received his Ph.D. degree from the Department of Computer Science, The University of Texas at Dallas, in 2021, and his B.E. degree from the School of Chemistry and Chemical Engineering, South China University of Technology, in 2015. He is currently an associate professor at the Institute of AI and Future Networks, Beijing Normal University, China, and also at the Guangdong Key Lab of AI and Multi-Modal Data Processing, Beijing Normal-Hong Kong Baptist University, China. He is a member of IEEE/ACM/CCF. He has published more than 120 peer-reviewed papers and has served as a reviewer for numerous international journals/conferences. His research interests include edge intelligence, IoT, distributed machine learning, and online/combinatorial optimization.



Wei Bao (Member, IEEE) received the BE degree in communications engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2009, the MASc degree in electrical and computer engineering from the University of British Columbia, Vancouver, Canada, in 2011, and the PhD degree in electrical and computer engineering from the University of Toronto, Toronto, Canada, in 2016. He is currently an associate professor with the School of Computer Science, the University of Sydney, Sydney, Australia. His research covers the area of network science, with particular emphasis on Internet of things, mobile computing, edge computing, and distributed learning. He received the Best Paper Awards in ACM International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems (MSWiM) in 2013 and 2019 and IEEE International Symposium on Network Computing and Applications (NCA), in 2016.



Min Zhou received the BS degree in automation from the University of Science and Technology of China, and the PhD degree from Industrial Systems Engineering and Management Department, National University of Singapore, respectively. She is currently with Huawei Co., Ltd., China. Her interests include pattern mining and machine learning, and their applications in sequence and graph data. Her several works related to graph learning and mining were published at top conferences, including KDD, WWW, ICDE, and SIGIR.

SUPPLEMENTARY MATERIALS

APPENDIX A

PROOF OF LEMMA 1

Proof: We first prove $\mathbf{Q}_t^n \preceq \frac{G}{\eta_t} \mathbf{1}, \forall t, \forall n$ by induction. From the initialization of the virtual queue, we have $\mathbf{Q}_1^n = \gamma_1 \mathbf{1} \preceq \frac{G}{\eta_1} \mathbf{1}, \forall n$. Then, suppose $Q_{\tau-1}^{n,c} \leq \frac{G}{\eta_{\tau-1}}, \forall n, \forall c$ for some $\tau > 1$, we now prove $Q_\tau^{n,c} \leq \frac{G}{\eta_\tau}, \forall n, \forall c$. From the virtual queue updating rule (19), we consider the following two cases:

- If $(1 - \eta_\tau)Q_{\tau-1}^{n,c} + [g_\tau^{n,c}(\mathbf{x}_\tau)]_+ > \gamma_\tau$, we have

$$\begin{aligned} Q_\tau^{n,c} &= (1 - \eta_\tau)Q_{\tau-1}^{n,c} + [g_\tau^{n,c}(\mathbf{x}_\tau)]_+ \\ &\leq (1 - \eta_\tau)Q_{\tau-1}^{n,c} + |g_\tau^{n,c}(\mathbf{x}_\tau)| \\ &\stackrel{(a)}{\leq} (1 - \eta_\tau)\frac{G}{\eta_{\tau-1}} + G \stackrel{(b)}{\leq} (1 - \eta_\tau)\frac{G}{\eta_\tau} + G = \frac{G}{\eta_\tau}, \end{aligned} \quad (44)$$

where (a) follows from $Q_{\tau-1}^{n,c} \leq \frac{G}{\eta_{\tau-1}}$ by induction and the bound on $|g_t^{n,c}(\mathbf{x})|$ in Assumption 3.1, and (b) is because of $\eta_t \in (0, 1)$ being non-increasing.

- Otherwise, $(1 - \eta_\tau)Q_{\tau-1}^{n,c} + [g_\tau^{n,c}(\mathbf{x}_\tau)]_+ \leq \gamma_\tau$, we have

$$Q_\tau^{n,c} = \gamma_\tau \leq \frac{G}{\eta_\tau}. \quad (45)$$

Combining the above two cases, we have proven by induction that $\mathbf{Q}_t^n \preceq \frac{G}{\eta_t} \mathbf{1}, \forall t, \forall n$.

Further noting that $\mathbf{Q}_1^n = \gamma_1 \mathbf{1}, \forall n$ from initialization and $\mathbf{Q}_t^n \succeq \gamma_t \mathbf{1}, \forall t > 1, \forall n$ from (19), we complete the proof. ■

APPENDIX B

PROOF OF LEMMA 2

Proof: From the virtual queue updating rule (19), we prove

$$\|\mathbf{Q}_t^n - \gamma_t \mathbf{1}\|^2 \leq \|(1 - \eta_t)\mathbf{Q}_{t-1}^n + [\mathbf{g}_t^n(\mathbf{x}_t)]_+ - \gamma_{t-1} \mathbf{1}\|^2 \quad (46)$$

for any n and $t > 1$, by considering the following two cases:

- If $(1 - \eta_t)Q_{t-1}^{n,c} + [g_t^{n,c}(\mathbf{x}_t)]_+ > \gamma_t$, we have

$$\begin{aligned} (Q_t^{n,c} - \gamma_t)^2 &\leq (Q_t^{n,c} - \gamma_{t-1})^2 \\ &= [(1 - \eta_t)Q_{t-1}^{n,c} + [g_t^{n,c}(\mathbf{x}_t)]_+ - \gamma_{t-1}]^2. \end{aligned} \quad (47)$$

- Otherwise, $(1 - \eta_t)Q_{t-1}^{n,c} + [g_t^{n,c}(\mathbf{x}_t)]_+ \leq \gamma_t$, we have

$$(Q_t^{n,c} - \gamma_t)^2 = 0 \leq [(1 - \eta_t)Q_{t-1}^{n,c} + [g_t^{n,c}(\mathbf{x}_t)]_+ - \gamma_{t-1}]^2. \quad (48)$$

Dividing both sides of (46) by 2, we have

$$\begin{aligned} \frac{1}{2}\|\mathbf{Q}_t^n - \gamma_t \mathbf{1}\|^2 &\leq \frac{1}{2}\|(1 - \eta_t)\mathbf{Q}_{t-1}^n + [\mathbf{g}_t^n(\mathbf{x}_t)]_+ - \gamma_{t-1} \mathbf{1}\|^2 \\ &= \frac{1}{2}\|(\mathbf{Q}_{t-1}^n - \gamma_{t-1} \mathbf{1}) + ([\mathbf{g}_t^n(\mathbf{x}_t)]_+ - \eta_t \mathbf{Q}_{t-1}^n)\|^2 \\ &= \frac{1}{2}\|\mathbf{Q}_{t-1}^n - \gamma_{t-1} \mathbf{1}\|^2 + \frac{1}{2}\|[\mathbf{g}_t^n(\mathbf{x}_t)]_+ - \eta_t \mathbf{Q}_{t-1}^n\|^2 \\ &\quad + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_t^n(\mathbf{x}_t)]_+ - \gamma_{t-1} \mathbf{1}^\top [\mathbf{g}_t^n(\mathbf{x}_t)]_+ \\ &\quad - \eta_t (\mathbf{Q}_{t-1}^n)^\top (\mathbf{Q}_{t-1}^n - \gamma_{t-1} \mathbf{1}). \end{aligned} \quad (49)$$

We now bound the RHS of (49). For the second term on the RHS of (49), we have

$$\begin{aligned} \|[\mathbf{g}_t^n(\mathbf{x}_t)]_+ - \eta_t \mathbf{Q}_{t-1}^n\| &\stackrel{(a)}{\leq} \|[\mathbf{g}_t^n(\mathbf{x}_t)]_+ + \eta_t \mathbf{Q}_{t-1}^n\| \\ &\stackrel{(b)}{\leq} \|\mathbf{g}_t^n(\mathbf{x}_t)\| + \|\eta_t \mathbf{Q}_{t-1}^n\| \stackrel{(c)}{\leq} \sqrt{C}(G + \frac{\eta_t G}{\eta_{t-1}}) \stackrel{(d)}{\leq} 2G\sqrt{C}, \end{aligned} \quad (50)$$

where (a) follows from $\|\mathbf{a} - \mathbf{b}\| \leq \|\mathbf{a} + \mathbf{b}\|, \forall \mathbf{a}, \mathbf{b} \succeq \mathbf{0}$, (b) is because of the triangle inequality, (c) follows from the bound on $|g_t^{n,c}(\mathbf{x})|$ in Assumption 3.1 and the virtual queue upper bound in (20), and (d) is because of η_t being non-increasing.

For the third term on the RHS of (49), we have

$$\begin{aligned} &(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_t^n(\mathbf{x}_t)]_+ \\ &= (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t)]_+ + (\mathbf{Q}_{t-1}^n)^\top ([\mathbf{g}_t^n(\mathbf{x}_t)]_+ - [\mathbf{g}_{t-1}^n(\mathbf{x}_t)]_+) \\ &\stackrel{(a)}{\leq} (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t)]_+ + \frac{G\sqrt{C}}{\eta_{t-1}} \|\mathbf{g}_t^n(\mathbf{x}_t) - \mathbf{g}_{t-1}^n(\mathbf{x}_t)\|, \end{aligned} \quad (51)$$

where (a) follows from $\mathbf{a}^\top \mathbf{b} \leq \|\mathbf{a}\| \|\mathbf{b}\|$, the virtual queue upper bound in (20), and $\|[\mathbf{a}]_+ - [\mathbf{b}]_+\| \leq \|\mathbf{a} - \mathbf{b}\|$.

For the last term on the RHS of (49), from the virtual queue lower bound in (20), we have

$$-\eta_t (\mathbf{Q}_{t-1}^n)^\top (\mathbf{Q}_{t-1}^n - \gamma_{t-1} \mathbf{1}) \leq 0. \quad (52)$$

Substituting (50)-(52) into (49) and rearranging the terms, we have

$$\begin{aligned} \Delta_{t-1}^n &= \frac{1}{2}\|\mathbf{Q}_t^n - \gamma_t \mathbf{1}\|^2 - \|\mathbf{Q}_{t-1}^n - \gamma_{t-1} \mathbf{1}\|^2 \\ &\leq (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t)]_+ - \gamma_{t-1} \mathbf{1}^\top [\mathbf{g}_t^n(\mathbf{x}_t)]_+ \\ &\quad + \frac{G\sqrt{C}}{\eta_{t-1}} \|\mathbf{g}_t^n(\mathbf{x}_t) - \mathbf{g}_{t-1}^n(\mathbf{x}_t)\| + 2G^2C. \end{aligned} \quad (53)$$

From (53) and noting that $\|\mathbf{g}_t^n(\mathbf{x}_t) - \mathbf{g}_{t-1}^n(\mathbf{x}_t)\| \leq \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\|$, we have (22). ■

APPENDIX C

PROOF OF LEMMA 3

Proof: In the objective function of $\mathbf{P}_t^n$, the first term $\langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x} - \mathbf{y}_{t-1}^n \rangle$ is affine, the second term $\alpha_{t-1} \|\mathbf{x} - \mathbf{y}_{t-1}^n\|^2$ is $2\alpha_{t-1}$ -strongly convex, and the third term $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x})]_+$ is convex (since the maximum and linear combination of convex functions are also convex). Therefore, the objective function of $\mathbf{P}_t^n$ is $2\alpha_{t-1}$ -strongly convex. Applying Lemma 5 to $\mathbf{P}_t^n$ and noting that $\mathbf{x}_t^n$ is the optimal solution to $\mathbf{P}_t^n$, we have

$$\begin{aligned} &\langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_t^n - \mathbf{y}_{t-1}^n \rangle + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \\ &\quad + \alpha_{t-1} \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 \\ &\leq \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n \rangle + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_{t-1}^*)]_+ \\ &\quad + \alpha_{t-1} \|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \alpha_{t-1} \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|^2. \end{aligned} \quad (54)$$

Summing (54) over n , noting that $[\mathbf{g}_{t-1}^n(\mathbf{x}_{t-1}^*)]_+ = \mathbf{0}$ from the definition of the dynamic online global solution benchmark $\mathbf{x}_t^*$ in (13), and then dividing both sides by N , we have

$$\begin{aligned} &\frac{1}{N} \sum_{n=1}^N [\langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_t^n - \mathbf{y}_{t-1}^n \rangle + \alpha_{t-1} \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 \\ &\quad + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+] \\ &\leq \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n \rangle \\ &\quad + \frac{\alpha_{t-1}}{N} \sum_{n=1}^N [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|^2]. \end{aligned} \quad (55)$$

We now bound the last term on the RHS of (55). We have

$$\begin{aligned}
& \sum_{n=1}^N [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|^2] \\
&= \sum_{n=1}^N [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2] \\
&+ \sum_{n=1}^N [\|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2 - \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|^2]. \quad (56)
\end{aligned}$$

For the last term on the RHS of (56), we have

$$\begin{aligned}
& \sum_{n=1}^N [\|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2 - \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|^2] \\
&\stackrel{(a)}{\leq} \sum_{n=1}^N \left[\sum_{k=1}^N p_t^{n,k} \|\mathbf{x}_t^* - \mathbf{x}_t^k\|^2 - \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|^2 \right] \\
&\stackrel{(b)}{\leq} \sum_{n=1}^N [\|\mathbf{x}_t^* - \mathbf{x}_t^n\|^2 - \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|^2] \\
&\stackrel{(c)}{\leq} \sum_{n=1}^N [\|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\|(\|\mathbf{x}_t^* - \mathbf{x}_t^n\| + \|\mathbf{x}_{t-1}^* - \mathbf{x}_t^n\|)] \\
&\stackrel{(d)}{\leq} 2NR\|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\|, \quad (57)
\end{aligned}$$

where (a) follows from $\mathbf{y}_t^n = \sum_{k=1}^N p_t^{n,k} \mathbf{x}_t^k$ in (3) and the Jensen's inequality; (b) is because of the doubly stochastic property $\sum_{n=1}^N p_t^{n,k} = 1$ of the edge weight matrix in (9), such that $\sum_{n=1}^N \sum_{k=1}^N p_t^{n,k} \|\mathbf{x}_t^* - \mathbf{x}_t^k\|^2 = \sum_{k=1}^N \|\mathbf{x}_t^* - \mathbf{x}_t^k\|^2$; (c) follows from $\|\mathbf{a}\|^2 - \|\mathbf{b}\|^2 \leq \|\mathbf{a} - \mathbf{b}\|(\|\mathbf{a}\| + \|\mathbf{b}\|)$, and (d) is because the feasible set $\mathcal{X}$ being bounded in (4).

Substituting (57) into (56) and then (55), we have (24). ■

APPENDIX D PROOF OF LEMMA 4

Proof: Rearranging terms of (24) in Lemma 3, we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \\
&\leq \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^* - \mathbf{x}_t^n \rangle \\
&- \frac{\alpha_{t-1}}{N} \sum_{n=1}^N \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 + 2R\alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \\
&+ \frac{\alpha_{t-1}}{N} \sum_{n=1}^N [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2]. \quad (58)
\end{aligned}$$

We now bound the first term on the RHS of (58). We have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^* - \mathbf{x}_t^n \rangle \\
&= \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), (\mathbf{x}_{t-1}^* - \mathbf{x}_{t-1}^n) + (\mathbf{x}_{t-1}^n - \mathbf{x}_t^n) \rangle \\
&\stackrel{(a)}{\leq} \frac{1}{N} \sum_{n=1}^N [f_{t-1}^n(\mathbf{x}_{t-1}^*) - f_{t-1}^n(\mathbf{x}_{t-1}^n) + D_1 \|\mathbf{x}_{t-1}^n - \mathbf{x}_t^n\|], \quad (59)
\end{aligned}$$

where (a) follows from the first-order condition for convexity, $\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\| \|\mathbf{b}\|$, and $\nabla f_t^n(\mathbf{x})$ being bounded in (5).

For the second term on the RHS of (59), we have

$$\begin{aligned}
& - \sum_{n=1}^N f_{t-1}^n(\mathbf{x}_{t-1}^n) \\
&\stackrel{(a)}{=} - \sum_{n=1}^N f_{t-1}(\mathbf{x}_{t-1}^n) + \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^N [f_{t-1}^k(\mathbf{x}_{t-1}^n) - f_{t-1}^k(\mathbf{x}_{t-1}^k)] \\
&\stackrel{(b)}{\leq} - \sum_{n=1}^N f_{t-1}(\mathbf{x}_{t-1}^n) + \frac{D_1}{N} \sum_{n=1}^N \sum_{k=1}^N \|\mathbf{x}_{t-1}^n - \mathbf{x}_{t-1}^k\| \\
&\stackrel{(c)}{\leq} - \sum_{n=1}^N f_{t-1}(\mathbf{x}_{t-1}^n) + 2D_1 \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\|, \quad (60)
\end{aligned}$$

where (a) follows from the definition of the global loss function $f_t(\mathbf{x})$ in (1); (b) is because $f_t^n(\mathbf{x})$ being convex, $\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\| \|\mathbf{b}\|$, and $\nabla f_t^n(\mathbf{x})$ being bounded in (5), such that $f_{t-1}^k(\mathbf{x}_{t-1}^n) - f_{t-1}^k(\mathbf{x}_{t-1}^k) \leq \langle \nabla f_{t-1}^k(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^n - \mathbf{x}_{t-1}^k \rangle \leq D_1 \|\mathbf{x}_{t-1}^n - \mathbf{x}_{t-1}^k\|$; and (c) follows from $\|\mathbf{x}_{t-1}^n - \mathbf{x}_{t-1}^k\| = \|(\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}) + (\bar{\mathbf{x}}_{t-1} - \mathbf{x}_{t-1}^k)\|$ and $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$.

Substituting (59) and (60) into (58), noting that $\frac{1}{N} \sum_{n=1}^N f_{t-1}^n(\mathbf{x}_{t-1}^n) = f_{t-1}(\mathbf{x}_{t-1}^*)$, and then rearranging terms, we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N [f_{t-1}(\mathbf{x}_{t-1}^n) - f_{t-1}(\mathbf{x}_{t-1}^*) + (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+] \\
&\leq \frac{D_1}{N} \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \mathbf{x}_t^n\| + \frac{2D_1}{N} \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| \\
&- \frac{\alpha_{t-1}}{N} \sum_{n=1}^N \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 + 2R\alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \\
&+ \frac{\alpha_{t-1}}{N} \sum_{n=1}^N [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2]. \quad (61)
\end{aligned}$$

We then bound the first term on the RHS of (61). We have

$$\begin{aligned}
& \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \mathbf{x}_t^n\| \\
&\stackrel{(a)}{\leq} \sum_{n=1}^n [\|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| + \|\bar{\mathbf{x}}_{t-1} - \mathbf{y}_{t-1}^n\| + \|\mathbf{y}_{t-1}^n - \mathbf{x}_t^n\|] \\
&\stackrel{(b)}{\leq} \sum_{n=1}^n [2\|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| + \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|], \quad (62)
\end{aligned}$$

where (a) follows from $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$; (b) is because of the definition of $\mathbf{y}_t^n$ in (3), the Jensen's inequality, and the doubly stochastic property $\sum_{n=1}^N p_t^{n,k} = 1$ of the edge weight matrix in (9), such that $\sum_{n=1}^N \|\bar{\mathbf{x}}_{t-1} - \mathbf{y}_{t-1}^n\| \leq \sum_{n=1}^N \sum_{k=1}^N p_t^{n,k} \|\bar{\mathbf{x}}_{t-1} - \mathbf{x}_{t-1}^k\| = \sum_{k=1}^N \|\bar{\mathbf{x}}_{t-1} - \mathbf{x}_{t-1}^k\|$.

Substituting (62) into (61), we have (25). ■

APPENDIX E AUXILIARY LEMMAS

Lemma 5. ([2], Lemma 2.8) Let $\mathcal{X}$ be a convex set. Let $f(\mathbf{x}) : \mathcal{X} \rightarrow \mathbb{R}$ be a 2α -strongly convex function with respect to a

norm $\|\cdot\|$, and $\mathbf{x}^* \in \arg \min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$ be the optimal solution of $f(\mathbf{x})$. Then, we have $f(\mathbf{x}^*) \leq f(\mathbf{x}) - \alpha \|\mathbf{x} - \mathbf{x}^*\|, \forall \mathbf{x} \in \mathcal{X}$.

Lemma 6. ([57], Proposition 1) Let $\Phi(t, s) = \mathbf{p}_s \mathbf{p}_{s+1} \cdots \mathbf{p}_t$ be the network transition matrix from time s to t , where $\mathbf{p}_t$ is the edge weight matrix at time t , with its (n, k) -th element being the edge weight $p_t^{n,k}$ from learner k to learner n . Under Assumption 4, each (n, k) element $\Phi(t, s)^{n,k}$ of $\Phi(t, s)$ converges to $\frac{1}{N}$ geometrically as follows

$$|\Phi(t, s)^{n,k} - \frac{1}{N}| \leq \rho \lambda^{t-s}, \quad \forall t \geq s \geq 1, \forall n, k \in \mathcal{N}, \quad (63)$$

where $\rho = \frac{2(1+w^{-B_0})}{1-w^{B_0}} \geq 2$, $\lambda = (1-w^{B_0})^{\frac{1}{B_0}} \in (0, 1)$, and $B_0 = (N-1)B$, with w and B being two constants defined in Assumptions 4.1 and 4.3, respectively.

Lemma 7. ([54], Lemma 4) The difference between the local decision $\mathbf{x}_t^n$ and the virtual global decision $\bar{\mathbf{x}}_t = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_t^n$ is bounded by

$$\begin{aligned} \|\mathbf{x}_t^n - \bar{\mathbf{x}}_t\| &\leq \rho \lambda^{t-2} \sum_{n=1}^N \|\mathbf{x}_1^n\| + \rho \sum_{s=1}^{t-2} \lambda^{t-s-2} \sum_{n=1}^N \|\mathbf{e}_s^n\| \\ &\quad + \frac{1}{N} \sum_{n=1}^N \|\mathbf{e}_{t-1}^n\| + \|\mathbf{e}_{t-1}^n\|, \end{aligned} \quad (64)$$

where $\mathbf{e}_t^n = \mathbf{x}_{t+1}^n - \mathbf{y}_t^n$ is the error between the local decision $\mathbf{x}_{t+1}^n$ and the local estimator $\mathbf{y}_t^n = \sum_{n=1}^N p_t^{n,k} \mathbf{x}_t^k$ in (3).

Lemma 8. ([1], Lemma 5) The following time series is upper bounded as

$$\sum_{t=1}^T \frac{1}{t^c} \leq \begin{cases} \frac{1}{1-c} T^{1-c}, & \text{if } 0 \leq c < 1; \\ \frac{1}{c-1}, & \text{if } c > 1. \end{cases} \quad (65)$$

APPENDIX F PROOF OF THEOREM 1

Proof: Summing (25) in Lemma 4 over $t = 2, \dots, T$, and noting that $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \geq 0$, we have

$$\begin{aligned} &\frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T f_{t-1}(\mathbf{x}_{t-1}^n) - \sum_{t=2}^T f_{t-1}(\mathbf{x}_{t-1}^*) \\ &\leq 2R \sum_{t=2}^T \alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{4D_1}{N} \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| \\ &\quad + \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T [D_1 \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\| - \alpha_{t-1} \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2] \\ &\quad + \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \alpha_{t-1} (\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2). \end{aligned} \quad (66)$$

We now bound the RHS of (66). To facilitate our analysis, we let $\mathbf{e}_t^n = \mathbf{x}_{t+1}^n - \mathbf{y}_t^n, \forall t$. For the second term on the RHS of (66), we have

$$\begin{aligned} &\frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| \\ &\stackrel{(a)}{\leq} \rho \sum_{n=1}^N \sum_{t=4}^T \lambda^{t-3} \|\mathbf{x}_1^n\| + \rho \sum_{n=1}^N \sum_{t=4}^T \sum_{s=1}^{t-3} \lambda^{t-3-s} \|\mathbf{e}_s^n\| \end{aligned}$$

$$\begin{aligned} &+ \frac{2}{N} \sum_{n=1}^N \sum_{t=4}^T \|\mathbf{e}_{t-2}^n\| + \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^3 \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| \\ &\stackrel{(b)}{\leq} \rho \sum_{n=1}^N \sum_{t=4}^T \lambda^{t-3} \|\mathbf{x}_1^n\| + \rho \sum_{n=1}^N \sum_{t=1}^{T-3} \|\mathbf{e}_t^n\| \sum_{s=0}^{T-t-3} \lambda^{t-3-s} \\ &\quad + \frac{2}{N} \sum_{n=1}^N \sum_{t=4}^T \|\mathbf{e}_{t-2}^n\| + 2R \\ &\stackrel{(c)}{\leq} \frac{\rho NR}{1-\lambda} + \frac{1}{N} \left(\frac{\rho N}{1-\lambda} + 2 \right) \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{e}_{t-1}^n\| + 2R \\ &\stackrel{(d)}{\leq} \frac{1}{N} \left(\frac{\rho N}{1-\lambda} + 2 \right) \sum_{n=1}^N \sum_{t=2}^T \frac{\kappa \alpha_{t-1}}{4} \|\mathbf{e}_{t-1}^n\|^2 \\ &\quad + \left(\frac{\rho N}{1-\lambda} + 2 \right) \sum_{t=2}^T \frac{1}{\kappa \alpha_{t-1}} + R \left(\frac{\rho N}{1-\lambda} + 2 \right), \end{aligned} \quad (67)$$

where (a) follows from applying (64) in Lemma 7 to $\|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\|$ for $t > 3$; (b) follows from the feasible set $\mathcal{X}$ being bounded in (4), and reorganizing $\sum_{t=4}^T \sum_{s=1}^{t-3} \lambda^{t-3-s} \sum_{n=1}^N \|\mathbf{e}_s^n\|$; (c) follows from $\|\mathbf{x}_1^n\| \leq R$ by initialization, and applying the geometric series summation formula, such that $\sum_{t=4}^T \lambda^{t-3} \leq \frac{1}{1-\lambda}$ and $\sum_{s=0}^{T-3-s} \lambda^s \leq \frac{1}{1-\lambda}$, and (d) is because $ab \leq \frac{\kappa a^2}{4} + \frac{b^2}{\kappa}, \forall \kappa > 0$.

For the third term on the RHS of (66), applying $ab \leq \frac{\kappa a^2}{4} + \frac{b^2}{\kappa}, \forall \kappa > 0$ again, we have

$$\|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\| \leq \frac{\kappa \alpha_{t-1}}{4} \|\mathbf{e}_{t-1}^n\|^2 + \frac{1}{\kappa \alpha_{t-1}}. \quad (68)$$

For the last term on the RHS of (66), we have

$$\begin{aligned} &\sum_{t=2}^T \alpha_{t-1} [\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2] \\ &\leq \alpha_1 \|\mathbf{x}_1^* - \mathbf{y}_1^n\|^2 + \sum_{t=2}^{T-1} (\alpha_t - \alpha_{t-1}) \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2 \\ &\stackrel{(a)}{\leq} \alpha_1 R^2 + \sum_{t=2}^{T-1} (\alpha_t - \alpha_{t-1}) R^2 = \alpha_{T-1} R^2, \end{aligned} \quad (69)$$

where (a) follows from $\mathcal{X}$ being bounded in (4).

Substituting (67), (68) and (69) into (66), and combining terms, we have

$$\begin{aligned} &\frac{1}{N} \sum_{n=1}^N \sum_{t=1}^{T-1} f_t(\mathbf{x}_t^n) - \sum_{t=1}^{T-1} f_t(\mathbf{x}_t^*) \\ &\leq 2R \sum_{t=2}^T \alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{D_1}{\kappa} \left(\frac{4\rho N}{1-\lambda} + 9 \right) \sum_{t=2}^T \frac{1}{\alpha_{t-1}} \\ &\quad + \frac{1}{N} \left[\frac{\kappa D_1}{4} \left(\frac{4\rho N}{1-\lambda} + 9 \right) - 1 \right] \sum_{n=1}^N \sum_{t=2}^T \alpha_{t-1} \|\mathbf{e}_{t-1}^n\|^2 \\ &\quad + \alpha_{T-1} R^2 + R D_1 \left(\frac{4\rho N}{1-\lambda} + 8 \right) \\ &\stackrel{(a)}{=} 2R \sum_{t=2}^T \alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{\varepsilon_1}{\kappa} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} + \alpha_{T-1} R^2 \\ &\quad + \frac{1}{N} \left(\frac{\kappa \varepsilon_1}{4} - 1 \right) \sum_{n=1}^N \sum_{t=2}^T \alpha_{t-1} \|\mathbf{e}_{t-1}^n\|^2 + R(\varepsilon_1 - D_1) \end{aligned}$$

$$\stackrel{(b)}{=} 2R \sum_{t=2}^T \alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{\varepsilon_1^2}{4} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} + \alpha_{T-1} R^2 + R(\varepsilon_1 - D_1), \quad (70)$$

where (a) follows from setting $\varepsilon_1 = D_1(\frac{4\rho N}{1-\lambda} + 9)$, and (b) follows from letting $\kappa = \frac{4}{\varepsilon_1}$.

Also, we have

$$f_t(\mathbf{x}_t^n) - f_t(\mathbf{x}_t^*) \stackrel{(a)}{\leq} \langle \nabla f_t(\mathbf{x}_t^n), \mathbf{x}_t^n - \mathbf{x}_t^* \rangle \stackrel{(b)}{\leq} RD_1, \quad (71)$$

where (a) is due to the convexity of $f_t(\mathbf{x})$, and (b) follows from $\langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\| \|\mathbf{y}\|$, the feasible set $\mathcal{X}$ being bounded in (4) and the gradient of the local loss function $\nabla f_t^n(\mathbf{x})$ being bounded in (5).

Combining (70) and $\frac{1}{N} \sum_{n=1}^N f_T(\mathbf{x}_T^n) - f_T(\mathbf{x}_T^*) \leq RD_1$ from (71), we have (26). ■

APPENDIX G PROOF OF THEOREM 3

Proof: Summing the penalized Lyapunov drift upper bound (22) in Lemma 2 over n , dividing both sides by N , and rearranging terms, we obtain the following lower bound on $\frac{1}{N} \sum_{n=1}^N (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+$, which is different from the 0 lower bound used in Theorem 1

$$\begin{aligned} & \frac{1}{N} \sum_{n=1}^N (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \\ & \geq \frac{\gamma_{t-1}}{N} \sum_{n=1}^N \mathbf{1}^\top [\mathbf{g}_t^n(\mathbf{x}_t^n)]_+ + \frac{1}{N} \sum_{n=1}^N \Delta_{t-1}^n \\ & \quad - \frac{G\sqrt{C}}{\eta_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| + 2G^2C. \end{aligned} \quad (72)$$

Substituting (72) into the per-slot performance bound of COLDQ in (25) of Lemma 4, and rearranging terms, we have

$$\begin{aligned} & \frac{\gamma_{t-1}}{N} \sum_{n=1}^N \mathbf{1}^\top [\mathbf{g}_t^n(\mathbf{x}_t^n)]_+ \\ & \leq \frac{G\sqrt{C}}{\eta_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| - \frac{1}{N} \sum_{n=1}^N \Delta_{t-1}^n \\ & \quad + \frac{1}{N} \sum_{n=1}^N \alpha_{t-1} \|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 + 2R\alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \\ & \quad + \frac{4D_1}{N} \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| + \frac{D_1}{N} \sum_{n=1}^N \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\| \\ & \quad + \frac{1}{N} \sum_{n=1}^N [f_{t-1}(\mathbf{x}_{t-1}^*) - f_{t-1}(\mathbf{x}_{t-1}^n)] + 2G^2C \\ & \stackrel{(a)}{\leq} \frac{G\sqrt{C}}{\eta_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| - \frac{1}{N} \sum_{n=1}^N \Delta_{t-1}^n \\ & \quad + 3\alpha_{t-1} R^2 + 6RD_1 + 2G^2C, \end{aligned} \quad (73)$$

where (a) follows from $\mathcal{X}$ being bounded in (4) and $f_{t-1}(\mathbf{x}_{t-1}^*) - f_{t-1}(\mathbf{x}_{t-1}^n) \leq RD_1$ similar to the proof of (71).

Dividing both sides of (73) by γ_{t-1} and summing over $t = 2, \dots, T$, we have

$$\begin{aligned} & \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \mathbf{1}^\top [\mathbf{g}_t^n(\mathbf{x}_t^n)]_+ \leq -\frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \frac{\Delta_{t-1}^n}{\gamma_{t-1}} \\ & \quad + \sum_{t=2}^T \frac{G\sqrt{C}}{\eta_{t-1}\gamma_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| \\ & \quad + 3R^2 \sum_{t=2}^T \frac{\alpha_{t-1}}{\gamma_{t-1}} + (6RD_1 + 2G^2) \sum_{t=2}^T \frac{1}{\gamma_{t-1}}. \end{aligned} \quad (74)$$

For the first term on the RHS of (74), we have

$$\begin{aligned} & -\sum_{t=2}^T \frac{\Delta_{t-1}^n}{\gamma_{t-1}} \stackrel{(a)}{=} -\frac{1}{2} \sum_{t=2}^T \left[\frac{\|\mathbf{Q}_t^n - \gamma_t \mathbf{1}\|^2}{\gamma_{t-1}} - \frac{\|\mathbf{Q}_{t-1}^n - \gamma_{t-1} \mathbf{1}\|^2}{\gamma_{t-1}} \right] \\ & \leq \frac{1}{2} \frac{\|\mathbf{Q}_1^n - \gamma_1 \mathbf{1}\|^2}{\gamma_1} + \frac{1}{2} \sum_{t=2}^T \left[\frac{\|\mathbf{Q}_t^n - \gamma_t \mathbf{1}\|^2}{\gamma_t} - \frac{\|\mathbf{Q}_t^n - \gamma_{t-1} \mathbf{1}\|^2}{\gamma_{t-1}} \right] \\ & \stackrel{(b)}{\leq} \frac{1}{2} \sum_{n=1}^N \frac{\|\mathbf{Q}_1^n - \gamma_1 \mathbf{1}\|^2}{\gamma_1} \stackrel{(c)}{=} 0, \end{aligned} \quad (75)$$

where (a) follows from the definition of Δ_{t-1} , (b) is because of γ_t being non-decreasing; and (c) follows from $\mathbf{Q}_1^n = \gamma_1 \mathbf{1}$.

Substituting $\|[\mathbf{g}_t^n(\mathbf{x}_t^n)]_+\| \leq \mathbf{1}^\top [\mathbf{g}_t^n(\mathbf{x}_t^n)]_+$ and (75) into the LHS and RHS of (74), respectively, and noting that $\|[\mathbf{g}_1^n(\mathbf{x}_1^n)]_+\| \leq \|\mathbf{g}_1^n(\mathbf{x}_1^n)\| \leq G\sqrt{C}$, we have (28). ■

APPENDIX H PROOF OF THEOREM 4

Proof: We first bound the per-slot hard constraint violation of each learner n as

$$\begin{aligned} & \|[\mathbf{g}_{t-1}^n(\mathbf{x}_{t-1}^n)]_+\| \\ & \stackrel{(a)}{\leq} \|[\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+\| + \|[\mathbf{g}_{t-1}^n(\mathbf{y}_{t-1}^n)]_+\| - \|[\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+\| \\ & \quad + \|[\mathbf{g}_{t-1}^n(\mathbf{x}_{t-1}^n)]_+\| - \|[\mathbf{g}_{t-1}^n(\mathbf{y}_{t-1}^n)]_+\| \\ & \stackrel{(b)}{\leq} \|[\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+\| + D_2\sqrt{C} [\|\mathbf{y}_{t-1}^n - \mathbf{x}_t^n\| + \|\mathbf{x}_{t-1}^n - \mathbf{y}_{t-1}^n\|] \\ & \stackrel{(c)}{\leq} \|[\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+\| + \beta D_2 \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 + \frac{CD_2}{4\beta} \\ & \quad + D_2\sqrt{C} \|\mathbf{x}_{t-1}^n - \mathbf{y}_{t-1}^n\|, \end{aligned} \quad (76)$$

where (a) follows from $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$; (b) is because $\|[\mathbf{a}]_+ - [\mathbf{b}]_+\| \leq \|\mathbf{a} - \mathbf{b}\|$, and $\mathbf{g}_t^n(\mathbf{x})$ being Lipschitz continuous in (7) of Assumption 3.2; and (c) follows from $ab \leq \beta a^2 + \frac{b^2}{4\beta}, \forall \beta > 0$.

Summing (76) over $n = 1, \dots, N$ and $t = 2, \dots, T$, and then dividing both sides by N , we have

$$\begin{aligned} & \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \|[\mathbf{g}_{t-1}^n(\mathbf{x}_{t-1}^n)]_+\| \\ & \leq \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \|[\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+\| + \frac{\beta D_2}{N} \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 \\ & \quad + \frac{D_2\sqrt{C}}{N} \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_{t-1}^n - \mathbf{y}_{t-1}^n\| + \frac{CD_2T}{4\beta}. \end{aligned} \quad (77)$$

We now bound the RHS of (77). For the first term on the RHS of (77), we have

$$\begin{aligned}
& \|[\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+\| \\
& \leq \mathbf{1}^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \stackrel{(a)}{\leq} \frac{(\mathbf{Q}_{t-1}^n)^\top}{\gamma_{t-1}} [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \\
& \stackrel{(b)}{\leq} \frac{1}{\gamma_{t-1}} \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^* - \mathbf{x}_t^n \rangle + \frac{\alpha_{t-1}}{\gamma_{t-1}} \|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 \\
& \stackrel{(c)}{\leq} \frac{RD_1}{\gamma_{t-1}} + \frac{\alpha_{t-1}R^2}{\gamma_{t-1}}, \tag{78}
\end{aligned}$$

where (a) is due to the virtual queue lower bound in (20); (b) follows from applying (54) in the proof of Lemma 3; and (c) follows from $\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\| \|\mathbf{b}\|$, and $\mathcal{X}$ and $\nabla f_t^n(\mathbf{x})$ being bounded in (4) and (5), respectively.

For the second term on the RHS of (77), we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 \\
& \stackrel{(a)}{\leq} \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \left[\frac{1}{\alpha_{t-1}} \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}_{t-1}^* - \mathbf{x}_t^n \rangle \right. \\
& \quad \left. + 2R\|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + (\|\mathbf{x}_{t-1}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}_t^* - \mathbf{y}_t^n\|^2) \right] \\
& \stackrel{(b)}{\leq} \sum_{t=2}^T \left[\frac{RD_1}{\alpha_{t-1}} + 2R\|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \right] + R^2, \tag{79}
\end{aligned}$$

where (a) follows from applying (24) of Lemma 3, and the proof of (b) is similar to the one for (c) in (78).

For the third term on the RHS of (77), we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_{t-1}^n - \mathbf{y}_{t-1}^n\| \\
& \stackrel{(a)}{\leq} \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T [\|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| + \|\bar{\mathbf{x}}_{t-1} - \mathbf{y}_{t-1}^n\|] \\
& \stackrel{(b)}{\leq} \frac{2}{N} \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| \\
& \stackrel{(c)}{\leq} \frac{2}{N} \left(\frac{\rho N}{1-\lambda} + 2 \right) \sum_{n=1}^N \sum_{t=2}^T \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\| + \frac{2\rho NR}{1-\lambda} + 4R \\
& \stackrel{(d)}{\leq} \frac{2}{N} \left(\frac{\rho N}{1-\lambda} + 2 \right) \sum_{n=1}^N \sum_{t=2}^T \left[\frac{\beta}{2} \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 + \frac{1}{2\beta} \right] \\
& \quad + \frac{2\rho NR}{1-\lambda} + 4R \\
& \stackrel{(e)}{\leq} \beta \left(\frac{\rho N}{1-\lambda} + 2 \right) \left(\sum_{t=2}^T \left[\frac{RD_1}{\alpha_{t-1}} + 2R\|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \right] + R^2 \right) \\
& \quad + \left(\frac{\rho N}{1-\lambda} + 2 \right) \frac{T}{\beta} + 2R \left(\frac{\rho N}{1-\lambda} + 2 \right), \tag{80}
\end{aligned}$$

where (a) is due to $\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|$, (b) follows from applying (62) in the proof of Lemma 4, (c) follows from applying step (c) of (67) in the proof of Theorem 1, (d) is due to applying $ab \leq \frac{\beta a^2}{2} + \frac{b^2}{2\beta}, \forall \beta > 0$, and (e) follows from applying (79).

Substituting (78)-(80) into (77), we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \|[\mathbf{g}_{t-1}^n(\mathbf{x}_{t-1}^n)]_+\| \\
& \stackrel{(a)}{\leq} RD_1 \sum_{t=2}^T \frac{1}{\gamma_{t-1}} + R^2 \sum_{t=2}^T \frac{\alpha_{t-1}}{\gamma_{t-1}} \\
& \quad + \varepsilon_2 \left[\beta \left(\sum_{t=2}^T \left[\frac{RD_1}{\alpha_{t-1}} + 2R\|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| \right] + R^2 \right) + \frac{T}{\beta} + 2R \right], \tag{81}
\end{aligned}$$

where (a) follows from letting $\varepsilon_2 = D_2[C + \sqrt{C}(\frac{\rho N}{1-\lambda} + 2)]$.

Combining (81) with $\|[\mathbf{g}_T^n(\mathbf{x}_T^n)]_+\| \leq \|\mathbf{g}_T^n(\mathbf{x}_T^n)\| \leq G\sqrt{C}$, and rearranging terms, we have (29). ■

APPENDIX I PROOF OF THEOREM 5

Proof: Replacing the dynamic online global solution benchmark $\{\mathbf{x}_t^*\}$ with the fixed offline benchmark $\mathbf{x}^*$ in (58) of the proof of Lemma 4, we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N (\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \\
& \leq \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}^* - \mathbf{x}_t^n \rangle - \frac{\alpha_{t-1}}{N} \sum_{n=1}^N \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 \\
& \quad + \frac{\alpha_{t-1}}{N} \sum_{n=1}^N [\|\mathbf{x}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}^* - \mathbf{y}_t^n\|^2]. \tag{82}
\end{aligned}$$

We now bound the first term on the RHS of (82). We have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), \mathbf{x}^* - \mathbf{x}_t^n \rangle \\
& = \frac{1}{N} \sum_{n=1}^N \langle \nabla f_{t-1}^n(\mathbf{x}_{t-1}^n), (\mathbf{x}^* - \mathbf{x}_{t-1}^n) + (\mathbf{x}_{t-1}^n - \mathbf{x}_t^n) \rangle \\
& \stackrel{(a)}{\leq} \frac{1}{N} \sum_{n=1}^N [f_{t-1}^n(\mathbf{x}^*) - f_{t-1}^n(\mathbf{x}_{t-1}^n) - \mu \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2] \\
& \quad + \frac{D_1}{N} \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \mathbf{x}_t^n\| \\
& \stackrel{(b)}{\leq} f_{t-1}(\mathbf{x}^*) - \sum_{n=1}^N f_{t-1}(\mathbf{x}_{t-1}^n) + 2D_1 \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| \\
& \quad + \frac{D_1}{N} \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \mathbf{x}_t^n\| - \frac{\mu}{N} \sum_{n=1}^N \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \\
& \stackrel{(c)}{\leq} f_{t-1}(\mathbf{x}^*) - \sum_{n=1}^N f_{t-1}(\mathbf{x}_{t-1}^n) + 4D_1 \sum_{n=1}^N \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| \\
& \quad + \frac{D_1}{N} \sum_{n=1}^N \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\| - \frac{\mu}{N} \sum_{n=1}^N \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2, \tag{83}
\end{aligned}$$

where (a) follows from $f_t^n(\mathbf{x})$ being μ -strongly convex in (30), $\langle \mathbf{a}, \mathbf{b} \rangle \leq \|\mathbf{a}\| \|\mathbf{b}\|$, and $\nabla f_t^n(\mathbf{x})$ being bounded in (5); (b) follows from applying (60) in the proof of Lemma 4; and (c) follows from applying (62) in the proof of Lemma 4.

Substituting (83) into (82) and summing the resulting inequality over $t = 2, \dots, T$. Further noting that $(\mathbf{Q}_{t-1}^n)^\top [\mathbf{g}_{t-1}^n(\mathbf{x}_t^n)]_+ \geq 0$, and rearranging terms, we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T f_{t-1}(\mathbf{x}_{t-1}^n) - \sum_{t=2}^T f_{t-1}(\mathbf{x}^*) \\
& \leq \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \left[4D_1 \|\mathbf{x}_{t-1}^n - \bar{\mathbf{x}}_{t-1}\| + D_1 \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\| \right. \\
& \quad \left. - \alpha_{t-1} \|\mathbf{x}_t^n - \mathbf{y}_{t-1}^n\|^2 \right] + \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \left[\alpha_{t-1} \|\mathbf{x}^* - \mathbf{y}_{t-1}^n\|^2 \right. \\
& \quad \left. - \alpha_{t-1} \|\mathbf{x}^* - \mathbf{y}_t^n\|^2 - \mu \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \right] \\
& \stackrel{(a)}{\leq} \frac{1}{N} \sum_{n=1}^N \sum_{t=2}^T \left[\alpha_{t-1} \|\mathbf{x}^* - \mathbf{y}_{t-1}^n\|^2 - \alpha_{t-1} \|\mathbf{x}^* - \mathbf{y}_t^n\|^2 \right. \\
& \quad \left. - \mu \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \right] + \frac{\varepsilon_1^2}{4} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} + R(\varepsilon_1 - D_1), \quad (84)
\end{aligned}$$

where (a) follows from applying (67) and (68) in the proof of Theorem 1.

For the first term on the RHS of (84), for any $t > 2$, we have

$$\begin{aligned}
& \sum_{n=1}^N \left[\alpha_{t-1} (\|\mathbf{x}^* - \mathbf{y}_{t-1}^n\|^2 - \|\mathbf{x}^* - \mathbf{y}_t^n\|^2) - \mu \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \right] \\
& = \sum_{n=1}^N \left[\alpha_{t-2} \|\mathbf{x}^* - \mathbf{y}_{t-1}^n\|^2 - \alpha_{t-1} \|\mathbf{x}^* - \mathbf{y}_t^n\|^2 \right] \\
& \quad + \sum_{n=1}^N \left[(\alpha_{t-1} - \alpha_{t-2}) \|\mathbf{x}^* - \mathbf{y}_{t-1}^n\|^2 - \mu \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \right] \\
& \stackrel{(a)}{\leq} \sum_{n=1}^N \left[\alpha_{t-2} \|\mathbf{x}^* - \mathbf{y}_{t-1}^n\|^2 - \alpha_{t-1} \|\mathbf{x}^* - \mathbf{y}_t^n\|^2 \right] \\
& \quad + \sum_{n=1}^N \left[(\alpha_{t-1} - \alpha_{t-2} - \mu) \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \right], \quad (85)
\end{aligned}$$

where (a) follows from the definition of the local estimator $\mathbf{y}_t^n$ in (3), the Jensen's inequality, and the doubly stochastic property $\sum_{n=1}^N p_t^{n,k} = 1$ of the edge weight matrix in (9), such that $\sum_{n=1}^N \|\mathbf{x}^* - \mathbf{y}_{t-1}^n\| \leq \sum_{n=1}^N \sum_{k=1}^N p_{t-1}^{n,k} \|\mathbf{x}^* - \mathbf{x}_{t-1}^k\| = \sum_{n=1}^N \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|$.

Substituting (85) for any $t > 2$ into (84), and noting that the first term on the RHS of (85) is telescoping, we have

$$\begin{aligned}
& \frac{1}{N} \sum_{n=1}^N \sum_{t=1}^{T-1} f_t(\mathbf{x}_t^n) - \sum_{t=1}^{T-1} f_t(\mathbf{x}^*) \\
& \stackrel{(a)}{\leq} \frac{1}{N} \sum_{n=1}^N \sum_{t=3}^T (\alpha_{t-1} - \alpha_{t-2} - \mu) \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 + \frac{\varepsilon_1^2}{4} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} \\
& \quad + \frac{1}{N} \sum_{n=1}^N \alpha_1 [\|\mathbf{x}^* - \mathbf{y}_1^n\|^2 + \|\mathbf{x}^* - \mathbf{y}_2^n\|^2] + R(\varepsilon_1 - D_1). \quad (86)
\end{aligned}$$

Combining (86) and $\frac{1}{N} \sum_{n=1}^N f_T(\mathbf{x}_T^n) - f_T(\mathbf{x}^*) \leq RD_1$ in (71), and from $\mathcal{X}$ being bounded in (4), we have (31). ■

APPENDIX J PROOF OF COROLLARY 1

1) *Network Dynamic Regret Bound:* Substituting $\alpha_t = t^a$, for any $a \in (0, 1)$, into the dynamic regret bound (26) in Theorem 1, we have

$$\begin{aligned}
& \text{REG}_d(T) \\
& \leq 2R \sum_{t=2}^T \alpha_{t-1} \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{\varepsilon_1^2}{4} \sum_{t=2}^T \frac{1}{\alpha_{t-1}} \\
& \quad + \alpha_{T-1} R^2 + \varepsilon_1 R \\
& \leq 2R \sum_{t=2}^T t^a \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{\varepsilon_1^2}{4} \sum_{t=1}^T \frac{1}{t^a} + R^2 T^a + \varepsilon_1 R \\
& \stackrel{(a)}{\leq} 2RT^a \sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{\varepsilon_1^2}{4(1-a)} T^{1-a} + R^2 T^a + \varepsilon_1 R \\
& \stackrel{(b)}{=} \mathcal{O}(T^{a+V_x}) + \mathcal{O}(T^{1-a}) + \mathcal{O}(T^a) + \mathcal{O}(1) \\
& \stackrel{(c)}{=} \mathcal{O}(T^{\frac{1+V_x}{2}}) + \mathcal{O}(T^{\frac{1+V_x}{2}}) + \mathcal{O}(1) \\
& = \mathcal{O}(T^{\frac{1+V_x}{2}}), \quad (87)
\end{aligned}$$

where (a) follows from applying Lemma 8, (b) is because $\sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| = \mathcal{O}(T^{V_x})$ in (17), and (c) follows from setting $a = \frac{1-V_x}{2}$ as in Corollary 1.

2) *Constraint Violation Bound with V_g :* Substituting $\alpha_t = t^a$, $\eta_t = \frac{1}{t^\varrho}$, and $\gamma_t = \epsilon t^\varrho$, for any $a \in (0, 1)$ and $\varrho > a + 1$, into the hard constraint violation bound (28) in Theorem 3, we have

$$\begin{aligned}
& \text{VIO}_h(T) \\
& \leq \sum_{t=2}^T \frac{G\sqrt{C}}{\eta_{t-1}\gamma_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| \\
& \quad + (6D_1R + 2G^2C) \sum_{t=2}^T \frac{1}{\gamma_{t-1}} + 3R^2 \sum_{t=2}^T \frac{\alpha_{t-1}}{\gamma_{t-1}} + G\sqrt{C} \\
& \leq \frac{G\sqrt{C}}{\epsilon} \sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, x \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| \\
& \quad + (6D_1R + 2G^2C) \frac{1}{\epsilon} \sum_{t=1}^T \frac{1}{t^\varrho} + \frac{3R^2}{\epsilon} \sum_{t=1}^T t^{a-\varrho} + G\sqrt{C} \\
& \stackrel{(a)}{\leq} \frac{G\sqrt{C}}{\epsilon} \sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, x \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| \\
& \quad + \frac{6D_1R + 2G^2C}{\epsilon(\varrho-1)} + \frac{3R^2}{\epsilon(\varrho-a-1)} + G\sqrt{C} \\
& \stackrel{(b)}{=} \mathcal{O}(T^{V_g}) + \mathcal{O}(1) \\
& = \mathcal{O}(T^{V_g}), \quad (88)
\end{aligned}$$

where (a) follows from applying Lemma 8, and (b) is because $\sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, x \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| = \mathcal{O}(T^{V_g})$ in (18).

3) *Constraint Violation Bound with V_x :* Substituting the same $\alpha_t = t^a$, $\eta_t = \frac{1}{t^\varrho}$, and $\gamma_t = \epsilon t^\varrho$, for any $a \in (0, 1)$ and $\varrho > a + 1$, into the hard constraint violation bound (29) in Theorem 4 instead, we have

$$\text{VIO}_h(T)$$

$$\begin{aligned}
&\leq \sum_{t=2}^T \frac{RD_1}{\gamma_{t-1}} + \sum_{t=2}^T \frac{\alpha_{t-1}R^2}{\gamma_{t-1}} + \varepsilon_2 \frac{T}{\beta} + \beta R^2 \varepsilon_2 \\
&\quad + 2\beta R \varepsilon_2 \sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \beta RD_1 \varepsilon_2 \sum_{t=2}^T \frac{1}{\alpha_{t-1}} \\
&\quad + 2R \varepsilon_2 + G\sqrt{C} \\
&\leq \frac{RD_1}{\epsilon} \sum_{t=2}^T \frac{1}{t^\varrho} + \frac{R^2}{\epsilon} \sum_{t=2}^T t^{a-\varrho} + \varepsilon_2 \frac{T}{\beta} + \beta R^2 \varepsilon_2 \\
&\quad + 2\beta R \varepsilon_2 \sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \beta RD_1 \varepsilon_2 \sum_{t=2}^T t^{-a} \\
&\quad + 2R \varepsilon_2 + G\sqrt{C} \\
&\stackrel{(a)}{\leq} \frac{RD_1}{\epsilon(\varrho-1)} + \frac{R^2}{\epsilon(\varrho-a-1)} + \varepsilon_2 \frac{T}{\beta} + \beta R^2 \varepsilon_2 \\
&\quad + 2\beta R \varepsilon_2 \sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| + \frac{RD_1 \varepsilon_2}{1-a} \beta T^{1-a} \\
&\quad + 2R \varepsilon_2 + G\sqrt{C} \\
&\stackrel{(b)}{=} \mathcal{O}\left(\frac{T}{\beta}\right) + \mathcal{O}(\beta) + \mathcal{O}(\beta T^{V_x}) + \mathcal{O}(\beta T^{1-a}) + \mathcal{O}(1) \\
&\stackrel{(c)}{=} \mathcal{O}(T^{\frac{3+V_x}{4}}) + \mathcal{O}(T^{\frac{1-V_x}{4}}) + \mathcal{O}(T^{\frac{1+3V_x}{4}}) \\
&= \mathcal{O}(T^{\frac{3+V_x}{4}}), \tag{89}
\end{aligned}$$

where (a) follows from applying Lemma 8; (b) is because $\sum_{t=2}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\| = \mathcal{O}(T^{V_x})$ in (17), and (c) follows from setting $a = \frac{1-V_x}{2}$ and $\beta = \frac{1-V_x}{4}$ as in Corollary 1. ■

APPENDIX K PROOF OF COROLLARY 5

1) *Network Static Regret Bound:* Substituting $\alpha_t = \mu t$ into the static regret bound (31) in Theorem 5, we have

$$\begin{aligned}
&\text{REG}_s(T) \\
&\leq \frac{1}{N} \sum_{n=1}^N \sum_{t=3}^T [\mu(t-1) - \mu(t-2) - \mu] \|\mathbf{x}^* - \mathbf{x}_{t-1}^n\|^2 \\
&\quad + \frac{\varepsilon_1^2}{4\mu} \sum_{t=1}^T \frac{1}{t} + 2\mu R^2 + R\varepsilon_1 \\
&= \frac{\varepsilon_1^2}{4\mu} \sum_{t=1}^T \frac{1}{t} + 2\mu R^2 + R\varepsilon_1 \\
&\leq \frac{\varepsilon_1^2}{4\mu} \log T + 2\mu R^2 + R\varepsilon_1 \\
&= \mathcal{O}(\log T) + \mathcal{O}(1) \\
&= \mathcal{O}(\log T). \tag{90}
\end{aligned}$$

2) *Constraint Violation Bound with V_g :* Substituting $\alpha_t = \mu t$, $\eta_t = \frac{1}{t^\varrho}$, and $\gamma_t = \epsilon t^\varrho$, for any $\varrho > 2$, into (28) in Theorem 3, with the dynamic online benchmark $\{\mathbf{x}_t^*\}$ replaced by the fixed offline benchmark $\mathbf{x}^*$, we have

$$\begin{aligned}
&\text{VIO}_h(T) \\
&\leq \sum_{t=2}^T \frac{G\sqrt{C}}{\eta_{t-1}\gamma_{t-1}} \max_{\mathbf{x} \in \mathcal{X}, n \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\|
\end{aligned}$$

$$\begin{aligned}
&+ (6RD_1 + 2G^2C) \sum_{t=2}^T \frac{1}{\gamma_{t-1}} + 3R^2 \sum_{t=2}^T \frac{\alpha_{t-1}}{\gamma_{t-1}} + G\sqrt{C} \\
&\leq \frac{G\sqrt{C}}{\epsilon} \sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, x \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| \\
&\quad + (6RD_1 + 2G^2C) \frac{1}{\epsilon} \sum_{t=1}^T \frac{1}{t^\varrho} + 3R^2 \frac{\mu}{\epsilon} \sum_{t=2}^T t^{1-\varrho} + G\sqrt{C} \\
&\stackrel{(a)}{\leq} \frac{G\sqrt{C}}{\epsilon} \sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, x \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| \\
&\quad + \frac{6RD_1 + 2G^2C}{\epsilon(\varrho-1)} + \frac{3R^2\mu}{\epsilon(\varrho-2)} + G\sqrt{C} \\
&\stackrel{(b)}{=} \mathcal{O}(T^{V_g}) + \mathcal{O}(1) \\
&= \mathcal{O}(T^{V_g}), \tag{91}
\end{aligned}$$

where (a) follows from Lemma 8, and (b) is because $\sum_{t=2}^T \max_{\mathbf{x} \in \mathcal{X}, x \in \mathcal{N}} \|\mathbf{g}_t^n(\mathbf{x}) - \mathbf{g}_{t-1}^n(\mathbf{x})\| = \mathcal{O}(T^{V_g})$ in (18).

3) *Constraint Violation Bound with V_x :* Substituting $\alpha_t = \mu t$, $\eta_t = \frac{1}{t^\varrho}$, and $\gamma_t = \epsilon t^\varrho$, for any $\varrho > 2$, into (29) in Theorem 4, with the dynamic online benchmark $\{\mathbf{x}_t^*\}$ replaced by the fixed offline benchmark $\mathbf{x}^*$, we have

$$\begin{aligned}
&\text{VIO}_h(T) \\
&\leq RD_1 \sum_{t=2}^T \frac{1}{\gamma_{t-1}} + R^2 \sum_{t=2}^T \frac{\alpha_{t-1}}{\gamma_{t-1}} + \varepsilon_2 \frac{T}{\beta} + \beta R^2 \varepsilon_2 \\
&\quad + \beta \varepsilon_2 \sum_{t=2}^T \frac{RD_1}{\alpha_{t-1}} + 2R \varepsilon_2 + G\sqrt{C} \\
&\leq \frac{RD_1}{\epsilon} \sum_{t=1}^T t^{-\varrho} + \frac{\mu R^2}{\epsilon} \sum_{t=1}^T t^{1-\varrho} + \varepsilon_2 \frac{T}{\beta} + \beta R^2 \varepsilon_2 \\
&\quad + \frac{\beta \varepsilon_2 RD_1}{\mu} \sum_{t=1}^T t^{-1} + 2R \varepsilon_2 + G\sqrt{C} \\
&\stackrel{(a)}{\leq} \frac{RD_1}{\epsilon(\varrho-1)} + \frac{\mu R^2}{\epsilon(\varrho-2)} + \varepsilon_2 \sqrt{T \log T} + R^2 \varepsilon_2 \sqrt{\frac{T}{\log T}} \\
&\quad + \frac{\varepsilon_2 RD_1}{\mu} \sqrt{T \log T} + 2R \varepsilon_2 + G\sqrt{C} \\
&= \mathcal{O}(\sqrt{T \log T}) + \mathcal{O}\left(\sqrt{\frac{T}{\log T}}\right) + \mathcal{O}(1) \\
&= \mathcal{O}(\sqrt{T \log T}), \tag{92}
\end{aligned}$$

where (a) follows from Lemma 8 and setting $\beta = \sqrt{\frac{T}{\log T}}$. ■

APPENDIX L EXPERIMENT DETAILS

In Table III, we summarize all algorithm parameters used for the experiment on time-varying constraints in Fig 1.

In Table IV, we summarize all algorithm parameters used for the experiment on fixed constraints in Fig 2.

TABLE III: Algorithm parameters for the experiment on time-varying constraints.

Algorithm	Parameters
COLDQ	$\alpha_t = t^{1/2}, \gamma = \epsilon t^2, \eta = 1/t^2, \epsilon = 0.5$
[27]	$\alpha_t = 0.5t^{1/2}, \gamma_t = t^{1/2+\epsilon}, \eta_t = t^{1/2}, \epsilon = 0.01$
[28]	$\beta = (2GD)^{-1}, G = 0.2, D = 10\sqrt{10}, V = 0.01$
[54]	$\alpha_t = 2/t^{1/2}, \gamma_t = 5/t^{1/2}, \beta_t = 0.2/t^{1/2}$

TABLE IV: Algorithm parameters for the experiment on fixed constraints.

Algorithm	Parameters
COLDQ	$\alpha_t = 2t^{1/2}, \gamma = \epsilon t^2, \eta = 1/t^2, \epsilon = 0.5$
[26]	$\alpha_t = 4/T^{1/2}, \gamma_t = 0.5T^{1/4}$
[27]	$\alpha_t = 2t^{1/2}, \gamma_t = 0.5t^{1/2+\epsilon}, \eta_t = 2t^{1/2}, \epsilon = 0.01$
[53]	$\eta_t = 1/T^c, \beta_t = 1/(apG^2T^c)$ $a = 5, p = 3, G = 1, c = 0.5$

In Table V, we summarize all algorithm parameters used for the convex logistic regression in Fig 3.

TABLE V: Algorithm parameters for the experiment on convex logistic regression.

Algorithm	Parameters
COLDQ	$\alpha_t = 1 \times 10^5, \gamma_t = \epsilon t^2, \eta_t = 1/t^2, \epsilon = 8$ $\sigma = 1 \times 10^{-6}, x_{\max} = 5 \times 10^{-3}, b = 3$
Idealized DL	$\alpha_t = 1 \times 10^5, x_{\max} = 5 \times 10^{-3}$
Quantized DL	$\alpha_t = 1 \times 10^5, \sigma = 1 \times 10^{-6}$ $x_{\max} = 5 \times 10^{-3}, b = 2$
[27]	$\alpha_t = 1 \times 10^5, \gamma_t = 60t^{1/2+\epsilon}, \eta_t = t^{1/2}, \epsilon = 0.01$ $\sigma = 1 \times 10^{-6}, x_{\max} = 5 \times 10^{-3}, b = 2$
[28]	$\alpha_t = 1 \times 10^5, \beta = 5 \times 10^2, V = 2.5 \times 10^{-3}$

In Table VI, we summarize all algorithm parameters used for the non-convex neural network training in Fig 4.

TABLE VI: Algorithm parameters for the experiment on non-convex neural network training.

Algorithm	Parameters
COLDQ	$\alpha_t = 50, \gamma_t = \epsilon t^2, \eta_t = 1/t^2, \epsilon = 0.02$ $\sigma = 1 \times 10^{-4}, x_{\max} = 0.1, b = 3$
Idealized DL	$\alpha_t = 50, x_{\max} = 0.1$
Quantized DL	$\alpha_t = 50, \sigma = 1 \times 10^{-4}, x_{\max} = 0.1, b = 2$
[27]	$\alpha_t = 50, \gamma_t = 0.05t^{1/2+\epsilon}, \eta_t = t^{1/2}, \epsilon = 0.01$ $\sigma = 1 \times 10^{-4}, x_{\max} = 0.1, b = 2$
[28]	$\alpha_t = 50, \beta = 1 \times 10^{-2}, V = 50$